Orthogonal Terrain-Following Coordinates for the Higher-Order Parabolic Equation: Wide-Angle Propagation Over Irregular Terrain With Discrete Transparent Boundaries

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***Abstract*—Parabolic equation (PE) solvers usually represent terrain by a staircase or by the Beilis–Tappert shift map. We map the propagation domain onto a rectangle by orthogonal terrain-following coordinates built from the tapered harmonic extension of the terrain. In these coordinates rational (Padé) propagators of arbitrary order apply without additional angle restrictions, and the discrete nonlocal transparent boundary condition (NLBC) remains exact, unlike with the shift map or conformal maps. The terrain enters the vertical Numerov scheme as a variable coefficient and a weight. A flux-conserving range step removes the amplitude drift caused by the range-dependent metric. A compact closure of the impedance condition has a reflection error bounded uniformly in the ground parameters. A discrete dispersion analysis proves local equivalence with the flat-terrain scheme. Applied to shift-map PEs, it reveals a slope error that does not vanish under grid refinement, even with an exact square root. On flat ground, these PEs reach 6–35° at an accuracy of 10^{-4} rad per wavelength, whereas the proposed scheme reaches $82°$, is 20 to 560 times cheaper, and has no maximum angle. Against exact solutions for beams reflected by conducting and impedance slopes at up to $70°$, its error is at most $3 \cdot 10^{-3}$, three orders of magnitude below that of the staircase and of the shift maps. Over real ground the staircase errs by 3–16% already on a $2°$ slope, because the steps impose the reflection coefficient of the wrong grazing angle. Absorbing layers must be tuned to the field's angular content and still leave errors up to three orders of magnitude above those of the NLBC. Two real-terrain examples are presented: one with weather model refractivity, and a mountain path over wet ground. On the mountain path, in V polarization, the staircase overestimates the field reflected by the slope by $10\,\mathrm{dB}$, twice the effect of the uncertainty in the ground parameters.**

***Index Terms*—Parabolic equation, irregular terrain, terrain-following coordinates, Padé approximation, nonlocal boundary conditions, impedance boundary condition, tropospheric propagation, numerical dispersion.**

I. Introduction

THE parabolic equation (PE) is the standard tool for predicting radio-wave propagation over irregular terrain in an inhomogeneous troposphere [1]. Finite-difference higher-order PE methods approximate the one-way propagator by a rational (Padé) function of the vertical operator [2]. Combined with a fourth-order vertical discretization [3], an automated choice of the order and of the grid [4], [5], and discrete nonlocal transparent boundary conditions (NLBCs) [6], [7], they resolve wide propagation angles on coarse grids and truncate the domain just above the region of interest.

Terrain, however, is usually still represented by a staircase: the field is set to zero below the terrain column by column. A staircase converges only at first order, scatters spuriously at every step, and for steep slopes demands a vertical step much smaller than the propagation angle requires. As shown below, the staircase error at wide angles does not decrease appreciably even with very fine grids. Over real ground the staircase also applies the impedance condition with the normal of the step rather than of the terrain, so the reflection coefficient is evaluated at the wrong grazing angle; near the Brewster angle this changes the reflected field by several decibels even on slopes of a few degrees.

The alternative is a change of variables that makes the terrain a coordinate line. The Beilis–Tappert shift map [8], combined with a phase factor, maps the narrow-angle (NA) PE over terrain exactly onto the PE over flat ground with an extra potential. It underlies TPEM, TEMPER and APM [9]–[11], as well as the acoustic PERUSE model [12] and its successors [13]–[15]. For the Helmholtz equation, however, the shear produces a mixed derivative, which the phase factor removes only at the paraxial level.

Recent shift-map finite-difference PEs: Recent work on finite-difference (FD) shift-map PEs falls into a few groups.

- *Wide-angle (WA) square roots with Crank–Nicolson (CN) marching.* These replace the square root by the Claerbout $[1/1]$ approximation, by the Greene approximation or by its Padé $[2/2]$ approximation, and march in range with CN [16]–[22]. Discretizations use three- or five-point differences in height and grids of $\Delta x = \Delta z = \lambda/2$ or $\Delta x \approx 5\lambda$, $\Delta z = \lambda/10$.
- *Reported limits.* On flat ground the reported angles are about $15°$ (NA), 36–$40°$ (Claerbout), $47°$ (Greene) and $55°$ (Padé $[2/2]$) [18], [19], [23]. Over terrain, the WA shift map is limited to slopes of 10–20° [16], [18], [21].
- *Wide-angle versions can be worse than NA.* The WA shift map is reported to be less accurate than the NA one on lee-side slopes steeper than about $5°$ [24], [25]. Gilbert [15] traced this to the vertical wavenumber in sheared coordinates being unphysical.
- *Required aperture.* The maximum angle required over terrain is the flat-ground aperture plus twice the slope [26].

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- *Domain truncation.* All these works truncate the domain from above with absorbing layers: Tukey or Hanning windows over the upper 30% of the domain [20], [27], a complex refractive index over the upper third [14], or perfectly matched layers [28], [29].
- *Accuracy analyses.* Grid-dispersion analyses of FD PEs [30] quantify the discretization error, but not the grid-independent error of the shift map.

Other approaches: The rotated PE [31], [32] is exact for piecewise-constant slopes but needs field interpolation at every slope break. Conformal maps preserve the Laplacian and have been combined with narrow-angle split-step Fourier solvers [12], [33], with bent optical waveguides [34], and with wide-angle rational PEs for water waves [35]. Local orthogonal transformations with exact one-way methods were developed for bounded acoustic waveguides [36]. Kampanis [37] used a terrain-fitting map that is the identity near the top so that an NLBC applies, but with a narrow-angle finite-element PE. To our knowledge, no method combines an exact (Helmholtz-level) treatment of the terrain, rational approximations of arbitrary order, and an exact discrete NLBC.

This paper provides such a method. Its contributions are as follows.

1) An explanation of why orthogonal coordinates are the natural setting for an arbitrary-order rational propagator with the NLBC (Section III): the shift map and conformal maps are not the identity near the upper boundary, whereas orthogonal coordinates can be, and keep the structure of the flat problem.
2) A construction of orthogonal terrain-following coordinates that coincide with Cartesian coordinates above a transition layer (Section IV).
3) The transformed one-way equation (Section V). Each factor of the rational propagator reduces to the Numerov scheme with a variable coefficient and one additional weight. A flux-conserving range step, written in a form that is the identity in the exterior, accounts for the range dependence of the metric.
4) A compact fourth-order closure of the Leontovich impedance condition on the terrain (Section VI-A). Its discrete reflection error is bounded uniformly in the ground parameters.
5) A discrete dispersion analysis (Section VII) showing local equivalence with the flat-terrain scheme and bounding the reflection error of the ground, and a demonstration that the discrete NLBC is unchanged (Section VIII).
6) A quantitative comparison with the NA and WA shift-map FD PEs (Section IX). The comparison uses the same discrete dispersion analysis: the grid-independent error floor of the shift map, the maximum angle reachable at a given accuracy, and the computational cost for a given angle. It is confirmed against exact solutions.
7) A comparison of the discrete NLBC with the absorbing layers used in the literature (Section X).
8) Validation against exact wide-angle solutions over conducting and impedance slopes, a comparison with the staircase over gentle impedance slopes, and two real-terrain paths. On the second, steep path over wet ground, the staircase error is compared with the effect of environmental uncertainty (Section XI).

II. Problem Statement

Consider the two-dimensional Helmholtz equation for a time-harmonic field $\psi(x,z)$ ($e^{-i\omega t}$ dependence)

$$\nabla^2\psi + k^2(x,z)\psi = 0 \tag{1}$$

in the domain $\Omega = \{z > T(x)\}$ above the terrain $T(x)$, where z is the height, $k = k_0 n$, and n is the modified refractive index, including the Earth flattening. The ground is either perfectly conducting, with $\psi = 0$ on $z = T(x)$ for horizontal (H) and $\partial_n\psi = 0$ for vertical (V) polarization, or it satisfies the Leontovich impedance condition [1]

$$\partial_n\psi + \alpha\psi = 0, \quad \alpha = ik_0\Delta_g, \quad \Delta_g = \begin{cases} \sqrt{\varepsilon - 1}, & \text{H}, \\ \sqrt{\varepsilon - 1}/\varepsilon, & \text{V}, \end{cases} \tag{2}$$

where ε is the complex relative permittivity of the ground and n is the normal directed into Ω. A plane wave incident at the grazing angle ϕ to the local ground is reflected with $R(\phi) = (\sin\phi - \Delta_g)/(\sin\phi + \Delta_g)$. Above some height the medium does not depend on x, and the domain is truncated there by an NLBC.

On flat ground let $L = \beta^{-2}\left(\partial_z^2 + k^2\right) - 1$ with a reference wavenumber β, and let $u = \psi e^{-i\beta x}$. The outgoing field obeys $u_x = i\beta(\sqrt{1+L} - 1)u$, and one range step is

$$e^{i\beta\Delta x\left(\sqrt{1+L}-1\right)} \approx R(L) = \prod_{l=1}^{p} \frac{1 + a_l L}{1 + b_l L}. \tag{3}$$

Each factor is a tridiagonal solve

$$\begin{aligned} b\,\delta^2 v + (1 + \alpha\delta^2)\left[\tau^2(1 + b\nu)v\right] \\ = a\,\delta^2 u + (1 + \alpha\delta^2)\left[\tau^2(1 + a\nu)u\right], \end{aligned} \tag{4}$$

where $\tau = \beta\Delta z$, $\alpha = 1/12$, $\nu = k^2/\beta^2 - 1$, and $\delta^2 u_j = u_{j+1} - 2u_j + u_{j-1}$. This is the Numerov–Padé scheme of [3]. The discrete NLBC [6] expresses the uppermost unknown of every factor through the history of the boundary values of all factors,

$$u^n_{N-1,l} - \sum_{l' \le l} Q^{(0)}_{ll'} u^n_{N,l'} = \sum_{m=1}^{n} \sum_{l'} Q^{(m)}_{ll'} u^{n-m}_{N,l'}. \tag{5}$$

The kernels $Q^{(m)}$ are the inverse z-transform of a matrix function of a_l, b_l and of the discrete exterior equation. They depend only on the exterior scheme and on the chain of factors.

III. Why Orthogonal Coordinates

A. Existing Transformations and the NLBC

The discrete NLBC (5) is derived for an exterior in which the discrete equation does not depend on range. A transformation that flattens the terrain must therefore become the identity at some height below the upper boundary. The existing transformations do not.

- *Shift map.* The substitution $\zeta = z - T(x)$ moves the whole column, including the upper boundary, with the terrain.

The transformed index $n^2(\zeta+T)-2\zeta T''$ and the phase factor $e^{ik_0T'\zeta}$ depend on range at every height and grow with ζ. This is why shift-map solvers truncate the domain with absorbing layers (Section X).

- *Conformal maps.* An analytic function that equals $w+c$ on an open set equals $w+c$ everywhere. A conformal map that flattens the terrain is therefore nowhere the identity, and the exterior index $n^2|F'|^2$ depends on range at every height. For a feature of horizontal scale ℓ, $|F'|^2-1$ decays like $e^{-2\pi\eta/\ell}$, so a $10\,\mathrm{km}$ hill still leaves a residual of order 10^{-1} at a height of $1\,\mathrm{km}$.

The staircase is compatible with the NLBC, but it does not treat the terrain as a boundary. We therefore look for a map that is the identity above a transition layer and, below it, keeps the structure of the flat problem that the rational propagator relies on. Orthogonal coordinates provide both.

B. Orthogonal Curvilinear Coordinates

Let $(x,z) = (X(\xi,\eta), Z(\xi,\eta))$ map the computational rectangle onto the propagation domain, with the terrain at $\eta=0$. Its tangent vectors are $\mathbf{a}_\xi = (X_\xi, Z_\xi)$, along the lines $\eta=$ const, and $\mathbf{a}_\eta = (X_\eta, Z_\eta)$, along the lines $\xi=$ const. The coordinates are orthogonal if $\mathbf{a}_\xi\cdot\mathbf{a}_\eta = 0$ everywhere. The arc length then has no cross term,

$$ds^2 = h_\xi^2\,d\xi^2 + h_\eta^2\,d\eta^2, \qquad h_\xi = |\mathbf{a}_\xi|, \quad h_\eta = |\mathbf{a}_\eta|. \tag{6}$$

The scale factors have a direct geometric meaning: a computational cell $\Delta\xi\times\Delta\eta$ is, to leading order, a rectangle with sides $h_\xi\Delta\xi$ along the level curve and $h_\eta\Delta\eta$ across it, rotated by the local slope of the level curve. Their ratio $r = h_\eta/h_\xi$ is the anisotropy of the cell; $r=1$ for a conformal map.

The gradient in these coordinates is $\nabla\psi = h_\xi^{-2}\mathbf{a}_\xi\psi_\xi + h_\eta^{-2}\mathbf{a}_\eta\psi_\eta$ and the area element is $h_\xi h_\eta\,d\xi\,d\eta$. The divergence theorem then gives $\nabla^2\psi = (h_\xi h_\eta)^{-1}\left[\partial_\xi(h_\eta h_\xi^{-1}\partial_\xi\psi) + \partial_\eta(h_\xi h_\eta^{-1}\partial_\eta\psi)\right]$, and multiplying (1) by $h_\xi h_\eta$ yields

$$\partial_\xi\left(r\partial_\xi\psi\right) + \partial_\eta\left(\frac{1}{r}\partial_\eta\psi\right) + h_\xi h_\eta k^2\psi = 0. \tag{7}$$

The unit normal to a level curve is $\mathbf{a}_\eta/h_\eta$, so the derivative along the normal is $\partial_n = h_\eta^{-1}\partial_\eta$.

Equation (7) has the three properties the method needs.

- *No mixed derivative.* In general coordinates the metric has an off-diagonal term, which produces $\partial_\xi\partial_\eta\psi$. For the shift map, for instance, $\psi_{xx} - 2s\psi_{x\zeta} + (1+s^2)\psi_{\zeta\zeta} + k^2\psi = 0$ with $s = T'$. With a mixed derivative the incident and reflected vertical wavenumbers of a plane wave at the boundary are no longer opposite, and no function of a single vertical operator is the exact propagator. In (7) the range and height derivatives are separated. After the Liouville scaling of Section V, the equation takes the form $\varphi_{\xi\xi} + B\varphi = 0$ with a vertical operator B, and the exact step is $\exp(i\Delta\xi\sqrt{B})$, which is approximated by (3) exactly as on flat ground.
- *Local boundary conditions.* Since $\partial_n = h_\eta^{-1}\partial_\eta$ on $\eta=0$, the Dirichlet, Neumann and impedance conditions remain conditions in η alone at a single node, and the vertical operator is symmetric with them. For frozen coefficients, $\eta\mapsto-\eta$ is a symmetry of (7), so the Dirichlet (Neumann) half-space solutions are odd (even) images, as on flat ground.
- *Local plane waves.* For frozen scale factors, $e^{i(\kappa_\xi\xi+\kappa_\eta\eta)}$ solves (7) if $(\kappa_\xi/h_\xi)^2 + (\kappa_\eta/h_\eta)^2 = k^2$, i.e. $\kappa_\xi = h_\xi k\cos\theta$ and $\kappa_\eta = h_\eta k\sin\theta$, where θ is the physical angle between the wave and the level curve. Locally the transformed problem is the flat problem with wavenumber $h_\xi k$ and vertical stretch r, and the grid $(\Delta\xi,\Delta\eta)$ acts as the physical grid $(h_\xi\Delta\xi, h_\eta\Delta\eta)$ aligned with the terrain. This is the basis of the dispersion analysis of Section VII.

Orthogonality leaves one family of coordinate lines free: once the level curves $\eta=$ const are chosen, the lines $\xi=$ const are their orthogonal trajectories and are determined uniquely. We use this freedom to make the level curves coincide with the terrain at $\eta=0$ and with horizontal lines above the transition layer (Section IV). The price, compared with a conformal map, is the anisotropy $r\neq1$, which enters the scheme only as a variable coefficient.

IV. Orthogonal Terrain-Following Coordinates

A. Level Curves

The coordinate lines $\eta=$ const are the curves $z = F(x,\eta) = \eta + z_{\mathrm{ref}} + u(x,\eta)$, where u is the harmonic extension of $b = T - z_{\mathrm{ref}}$, tapered to zero:

$$\hat u(\kappa,\eta) = \hat b(\kappa)\,w(\eta/L)\,e^{-|\kappa|\eta}, \tag{8}$$

$$w(t) = \begin{cases} 1 - t^3(10 - 15t + 6t^2), & 0\le t\le 1,\\ 0, & t>1. \end{cases}$$

Here $\hat{}$ denotes the Fourier transform in x, and L is the height of the transition layer. The construction has four properties.

- *Exact boundary.* $F(x,0) = T(x)$.
- *Identity above the layer.* $F = \eta + z_{\mathrm{ref}}$ for $\eta\ge\eta_\ell = L$: the coordinates are Cartesian, up to a constant shift, above the transition layer.
- *Near-conformality near the ground.* Near the ground u is harmonic, so a feature of horizontal scale ℓ decays over a height of about ℓ. Small-scale relief stays near the ground, as with a conformal map, and only the large-scale relief reaches the taper.
- *Regularity.* With $u = wU$, the map is regular when $F_\eta = 1 + wU_\eta + w'U/L > 0$. For $z_{\mathrm{ref}} = \max T$ we have $U\le0$ (maximum principle) and $w'\le0$, so the taper term is non-negative for every L, and $F_\eta \ge 1 - \max|U_\eta|$ is a bound on the smoothed slopes. For $z_{\mathrm{ref}} = \overline{T}$ the bound acquires the term $-\frac{15}{8}\max(T - z_{\mathrm{ref}})/L$.

The terrain samples are filtered with a Gaussian of width σ to remove the corners of the piecewise-linear profile, and are extended evenly at $x=0$. The even extension makes $u_x(0,\eta) = 0$, so the starting line $\xi=0$ is the vertical line $x=0$ and the antenna pattern is prescribed in height as usual.

B. Orthogonal Trajectories and Metrics

The level curves fix η. It remains to find the lines $\xi=$ const and the scale factors. A node (ξ,η) is mapped to

$$x = X(\xi,\eta), \qquad z = F\big(X(\xi,\eta),\eta\big),$$

where X is the abscissa of the line $\xi = \text{const}$ at the level η; the second relation says that the node lies on the level curve η, so that $Z(\xi,\eta) = F(X(\xi,\eta),\eta)$ in the notation of Section III. Since F is known from (8), the only unknown is X. The tangents to the two coordinate lines are

$$\mathbf{a}_\xi = (X_\xi, Z_\xi) = P\,(1, F_x),$$
$$\mathbf{a}_\eta = (X_\eta, Z_\eta) = (X_\eta,\, F_x X_\eta + F_\eta),$$

by the chain rule, where $P = X_\xi$ and F_x, F_η are taken at $x = X$.

Trajectories. Setting the scalar product of the two tangents to zero gives an ordinary differential equation for each line $\xi = \text{const}$:

$$X_\eta = G(X,\eta) \equiv -\frac{F_x F_\eta}{1+F_x^2}, \qquad X(\xi,\eta_\ell) = \xi. \tag{9}$$

Above the layer $G = 0$ and the lines are vertical, so they are labeled by their abscissa there. Integrating (9) downward from η_ℓ to 0 traces each line through the layer, crossing every level curve at a right angle. The trajectories are integral curves of the non-vanishing field $\nabla\eta$ and never cross.

Scale factors. The scale factors are the lengths of the two tangents. With $X_\eta = G$ we have $F_x X_\eta + F_\eta = F_\eta/(1+F_x^2)$, and

$$h_\eta = \frac{F_\eta}{\sqrt{1+F_x^2}}, \quad h_\xi = P\sqrt{1+F_x^2}, \quad P_\eta = G_x(X,\eta)P, \tag{10}$$

with $P(\xi,\eta_\ell) = 1$. The equation for P follows by differentiating (9) in ξ. Both factors have a simple meaning. Neighboring level curves are $F_\eta\,d\eta$ apart vertically; projected on the normal, which is tilted by $\arctan F_x$, this gives h_η. It depends on F at the node only. Neighboring trajectories are $P\,d\xi$ apart horizontally; converted to arc length along the level curve, this gives h_ξ. Through P, which accumulates the convergence ($P<1$) or divergence ($P>1$) of the trajectories on their way down, h_ξ depends on the whole path above the node. As a check, $h_\xi h_\eta = PF_\eta$ equals the Jacobian of the map, and the anisotropy is $r = F_\eta/[P(1+F_x^2)]$.

Computation. The function $\hat{b}$ is computed once by FFT. On each level, one inverse FFT of $\hat{u}$ and of its products with $i\kappa$, $-\kappa^2$ and ∂_η gives F_x, F_η, F_{xx} and $F_{x\eta}$ on a fine uniform grid in x; they are interpolated to the current X to form G and G_x. Equations (9)–(10) are integrated downward with the fourth-order Runge–Kutta method for all columns at once, on coarse levels, and the metrics are then interpolated to the computational nodes with cubic splines. The map takes seconds to build. Orthogonality of the computed grid was verified to $2\cdot10^{-4}$, the level of the finite-difference check itself. Fig. 1 shows the resulting grids.

V. Transformed One-Way Equation

A. Liouville Transformation

The range dependence of r enters (7) through $\partial_\xi(r\partial_\xi)$. Setting $\psi = r^{-1/2}\varphi$ and using $\partial_\xi(r\partial_\xi(r^{-1/2}\varphi)) = r^{1/2}\varphi_{\xi\xi} - (r^{1/2})_{\xi\xi}\varphi$, we obtain

$$\varphi_{\xi\xi} + B\varphi = 0, \tag{11}$$
$$B\varphi = r^{-1/2}\partial_\eta\left(\frac{1}{r}\partial_\eta\left(r^{-1/2}\varphi\right)\right) + \left(k^2 h_\xi^2 - V\right)\varphi,$$

with $V = (r^{1/2})_{\xi\xi}/r^{1/2}$, a term of relative size $(\lambda/\ell)^2$. The operator B is symmetric in $L^2(d\eta)$. Dirichlet and Neumann conditions are invariant. Since $\partial_n = h_\eta^{-1}\partial_\eta$ on $\eta = 0$, the impedance condition (2) becomes the Robin condition

$$\partial_\eta\psi + \gamma\psi = 0, \qquad \gamma(\xi) = \alpha\, h_\eta(\xi,0), \tag{12}$$

with a coefficient that varies along the ground. The scheme below is written for ψ, so (12) is imposed on ψ directly.

B. Flux-Conserving Range Step

One-way equation. If B does not depend on ξ, the outgoing factor of (11) is $\varphi_\xi = i\sqrt{B}\varphi$. If it does, we seek $\varphi_\xi = i\Lambda\varphi$. Differentiating once more and substituting into (11) gives the Riccati equation $\Lambda^2 - i\Lambda_\xi = B$. Its expansion in the slowness of the range variation is $\Lambda = \sqrt{B} + \frac{i}{4}B^{-1}B_\xi + \ldots$, where at first order B and B_ξ are treated as commuting. Hence

$$\varphi_\xi = i\sqrt{B}\varphi - \tfrac{1}{4}B^{-1}B_\xi\varphi + \ldots, \tag{13}$$

where the dots denote second-order terms and commutators $[B, B_\xi]$. The second term is the logarithmic derivative $\partial_\xi \ln B^{1/4}$; in the scalar case (13) is solved by the WKB form $\varphi = B^{-1/4}\exp(i\int\sqrt{B}\,d\xi)$.

Energy flux. The time-averaged power through a line $\xi = \text{const}$ is proportional to $\operatorname{Im}\int\bar{\psi}\,\partial_n\psi\,dl$ with $\partial_n = h_\xi^{-1}\partial_\xi$ and $dl = h_\eta\,d\eta$. Since $\psi = r^{-1/2}\varphi$ and $r^{-1/2}$ is real, this equals

$$\mathcal{F} = \operatorname{Im}\int\bar{\varphi}\,\varphi_\xi\,d\eta = \operatorname{Re}\langle\varphi, \sqrt{B}\varphi\rangle = \|B^{1/4}\varphi\|^2, \tag{14}$$

where the second equality uses (13), and only the propagating part of the spectrum, on which $\sqrt{B} > 0$, contributes. With $\chi = B^{1/4}\varphi$ the transport term cancels, $\chi_\xi = i\sqrt{B}\chi + \ldots$, so χ is advanced by the exact exponential and $\mathcal{F} = \|\chi\|^2$ is conserved. The plain step $\varphi_{n+1} = e^{i\Delta\xi\sqrt{B}}\varphi_n$ drops the transport term and conserves $\|\varphi\|^2$ instead.

Why the term matters here. For a local plane wave at angle θ to the marching direction, $\sqrt{B} \approx kh_\xi\cos\theta$, so $B^{1/4} \propto (\cos\theta)^{1/2}$: the flux is the squared amplitude times $\cos\theta$, and when θ changes from θ_1 to θ_2 the amplitude must change by the WKB factor $(\cos\theta_1/\cos\theta_2)^{1/2}$. In the usual PE, B depends on range only through n^2, which varies over scales $\ell \gg \lambda$, and the term is neglected. In the proposed coordinates the lines rotate with height, so a beam crossing the transition layer changes its angle to the local marching direction by $O(1)$. The plain step then loses amplitude, by 2–6% over a few hundred meters for beams reflected at 40–50° in the tests of Section XI-A, in agreement with the WKB factor. Above the layer $B_\xi = 0$ and the term vanishes.

Discrete step. We use the flux-conserving step [38], [39]: map to χ with $B_n^{1/4}$, advance with the exponential at the

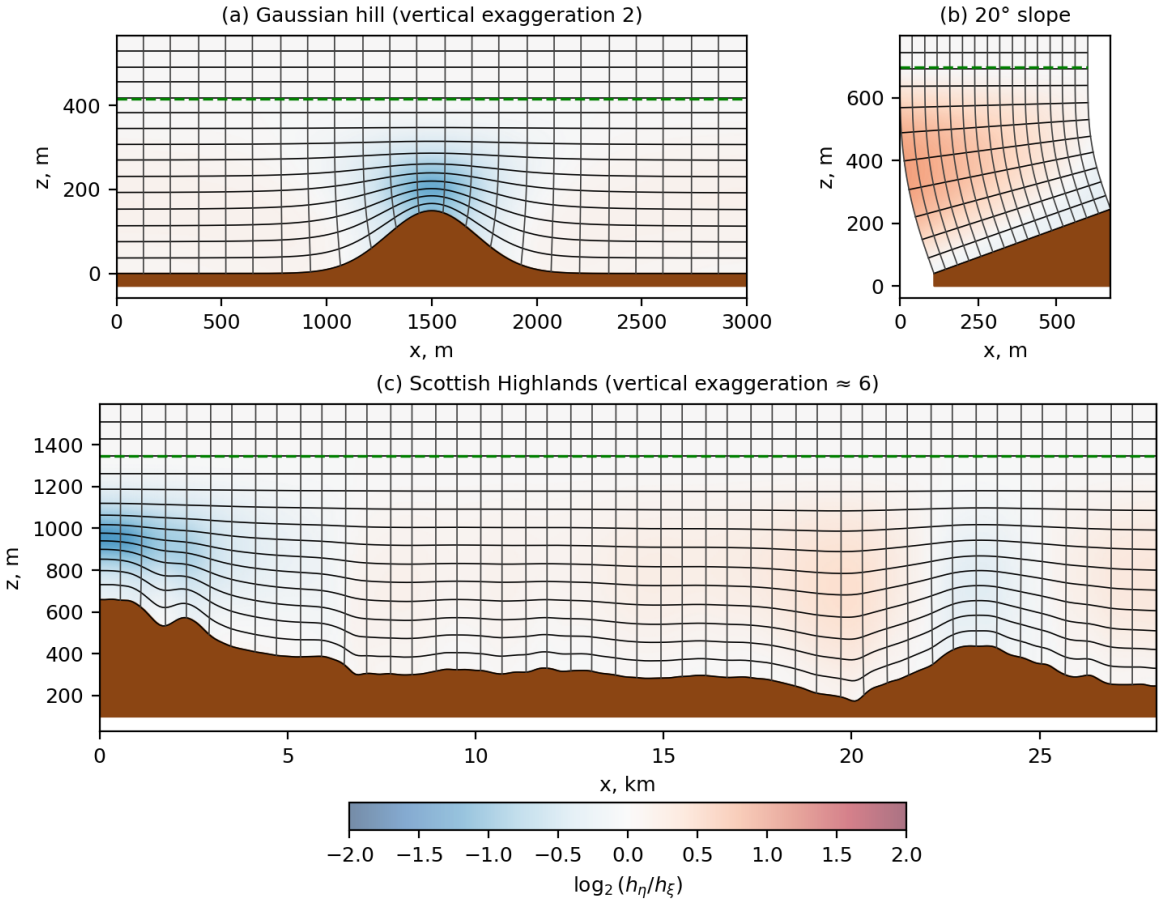


Fig. 1. Orthogonal terrain-following coordinates. Black: level curves $\eta = \text{const}$; grey: trajectories $\xi = \text{const}$; color: $\log_2(h_\eta/h_\xi)$; dashed: top of the transition layer, above which the coordinates are Cartesian. (a) Gaussian hill, $z_{\text{ref}} = \overline{T}$, $L = 400\,\text{m}$. (b) 20° slope without the even extension at $x = 0$; the starting trajectory is then curved. (c) Scottish Highlands profile of Section XI-E. Panels (a) and (c) are vertically exaggerated, so the right angles appear distorted.

midpoint, and map back with $B_{n+1}^{-1/4}$. Since the middle factor commutes with $B_{n+1/2}^{\pm 1/4}$, it can be written as

$$\varphi_{n+1} = \left[B_{n+1}^{-1/4} B_{n+1/2}^{1/4}\right] e^{i\Delta\xi\sqrt{B_{n+1/2}}} \left[B_{n+1/2}^{-1/4} B_n^{1/4}\right] \varphi_n = B_{n+1}^{-1/4} e^{i\Delta\xi\sqrt{B_{n+1/2}}} B_n^{1/4} \varphi_n. \quad (15)$$

Each bracket is the identity wherever its two columns coincide, in particular in the whole exterior. The middle factor is approximated by (3) with $L = B/\beta^2 - 1$.

The factors $(1+L)^{\pm 1/4}$ are approximated by rational functions. Their Padé approximants have poles and zeros on the branch cut $L < -1$, i.e., inside the evanescent spectrum. There a pair of factors evaluated at slightly different columns can amplify a spurious mode; we observed a 14% localized error after $600\,\text{m}$. Following [40], we take the $[q/q]$ Padé approximant of $(1+Y)^{1/4}$ with $Y = (1+L)e^{-i\gamma} - 1$ and $\gamma = 45°$. Rewritten as factors $(1+cL)/(1+dL)$, its poles and zeros lie off the real axis. The constant prefactor cancels within each pair, and only the logarithmic derivative matters, which for $q = 2$ is accurate to 5% up to 45°. Since the correction itself is a few percent, the resulting error is of order 10^{-3}.

C. Factors in Terms of the Physical Field

Let $\varphi = r^{1/2}\psi$, $m = r^2$ and $\nu = (k^2 h_\xi^2 - V)/\beta^2 - 1$. Inserting (11) into $\beta^2(1+cL)\varphi = \beta^2(1-c)\varphi + cB\varphi$ and multiplying by $r^{3/2}$ gives

$$r^{3/2}\beta^2(1+cL)\varphi = c\,r\,\partial_\eta(r^{-1}\partial_\eta\psi) + \beta^2 m(1+c\nu)\psi.$$

Every factor of (15) is therefore the flat scheme (4), with ∂_η^2 replaced by the operator $r\partial_\eta r^{-1}\partial_\eta$ with the variable coefficient $r = h_\eta/h_\xi$, and with an additional weight m. The factors of (15) are evaluated at the columns ξ_n, $\xi_{n+1/2}$ and ξ_{n+1}, and so is the substitution $\psi = r^{-1/2}\varphi$. When the scheme passes from factors at a column ξ_a to factors at the next column ξ_b, φ is unchanged, so ψ is multiplied by $(r_a/r_b)^{1/2} = (m_a/m_b)^{1/4}$, where $m_a = m(\xi_a, \eta)$ and $m_b = m(\xi_b, \eta)$.

VI. Discretization

On the uniform grid $\eta_j = j\Delta\eta$ each factor reads

$$c\,\delta_r^2 v + (1+\alpha\delta_r^2)\left[\tau^2 m(1+c\nu)v\right] = a\,\delta_r^2 u + (1+\alpha\delta_r^2)\left[\tau^2 m(1+a\nu)u\right] \quad (16)$$

with $\tau = \beta\Delta\eta$ and $\delta_r^2 u_j = r_j\left[(u_{j+1}-u_j)/r_{j+1/2} - (u_j - u_{j-1})/r_{j-1/2}\right]$. For

Algorithm 1: one range step

1: interpolate (ν, m, r) at ξ_n, $\xi_{n+1/2}$, ξ_{n+1}
2: $\psi \leftarrow C_1\psi$ (first bracket of (15); rows below η_ℓ plus a margin)
3: **for** $l = 1, \dots, p$ **do**
4: solve (16) with (a_l, b_l) at $\xi_{n+1/2}$ and the NLBC row (5)
5: **end for**
6: $\psi \leftarrow C_2\psi$ (second bracket of (15))

$g = \ln r$ and $D = r\partial r^{-1}\partial$, a Taylor expansion gives $\delta_r^2/\Delta\eta^2 = D + \frac{\Delta\eta^2}{12}D^2 + \Delta\eta^2 E + O(\Delta\eta^4)$, where $E\psi = \frac{1}{24}\left[(g'^2 + g'')\psi'' + (g''' + g'g'' - g'^3)\psi'\right]$. The Numerov weight cancels the D^2 term. The scheme is thus fourth-order in the wave part, with a second-order error of relative size $(\Delta\eta/\ell_\eta)^2/24$, negligible since $\ell_\eta \gtrsim L$.

At the ground the Dirichlet unknown is eliminated from the system rather than imposed by a unit row; a unit row next to entries of order $\alpha\tau^2$ is satisfied by a pivoting solver only up to amplified round-off. The Neumann condition uses a mirror node. The impedance closure is described in Section VI-A. The coefficients ν, m, r are tabulated on coarse columns, only below the Cartesian region, and are interpolated linearly in range; the main chain is evaluated at the midpoint. One step (Algorithm 1) costs $p + 4q$ tridiagonal solves, and the $4q$ flux solves are restricted to the transition layer.

A. Impedance Ground

For the Robin condition (12) a ghost node u_{-1} is added to the row $j = 0$ of (16), and the system is closed by

$$(u_1 - u_{-1}) + \frac{g}{3}(u_{-1} + 4u_0 + u_1) = 0, \qquad g = \gamma\Delta\eta. \quad (17)$$

This is (12) written with the compact (Padé) pair $(u_1 - u_{-1})/(2\Delta\eta) = \psi_0' + \frac{\Delta\eta^2}{6}\psi_0''' + \dots$ and $(u_{-1} + 4u_0 + u_1)/6 = \psi_0 + \frac{\Delta\eta^2}{6}\psi_0'' + \dots$ [41]. The $\Delta\eta^2$ terms combine into $\partial_\eta^2(\psi' + \gamma\psi)$. For frozen coefficients the equation turns ∂_η^2 into an operator along the ground, which annihilates (12), so the closure is fourth-order accurate. Relation (17) expresses u_{-1} through u_0 and u_1, so the system stays tridiagonal. The coefficients of row 0 at the ghost node are extrapolated quadratically, and the same closure is used in all $p + 4q$ factors of a step. The usual closure is the central difference $(u_1 - u_{-1})/(2\Delta\eta) + \gamma u_0 = 0$, as in the discrete mixed Fourier transform of split-step solvers [42]. We call it the mirror closure; it is second-order accurate.

VII. Discrete Dispersion Analysis

Freeze h_ξ, h_η and k, and consider a plane wave at angle θ to the local marching direction: $k_z = k\sin\theta$, computational wavenumber $q = k_z h_\eta$, $\psi_j^n = \lambda^n e^{iqj\Delta\eta}$. The flux factors cancel. With $s = \sin^2(q\Delta\eta/2)$, the symbol of (16) is

$$L_h(q) = \nu - \frac{4}{\tau^2 r^2}\frac{s}{1 - 4\alpha s}, \qquad \lambda = R\left(L_h(q)\right), \quad (18)$$

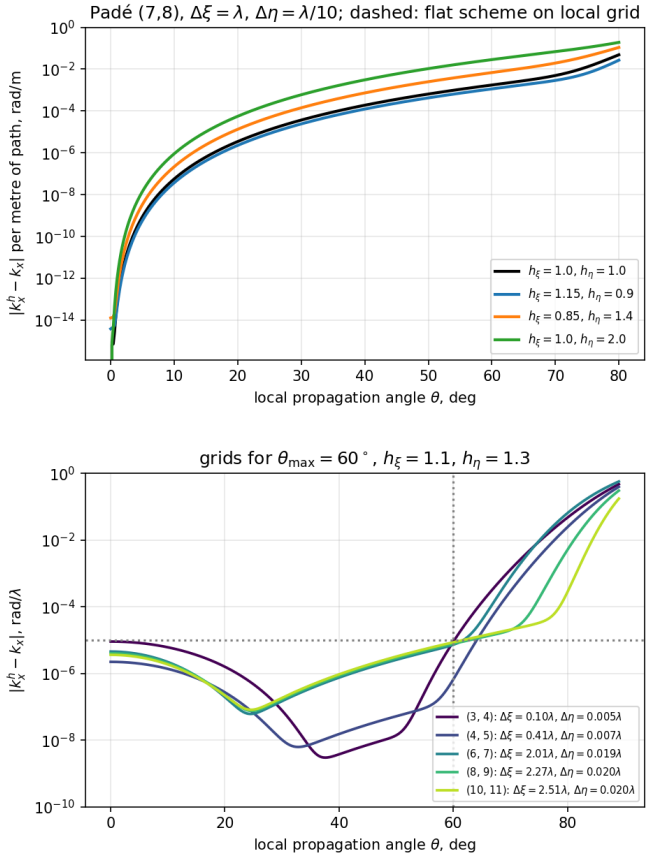


Fig. 2. Top: dispersion error for several metric values on one grid (solid) and of the flat scheme on the local effective grid (dashed); they agree to $4\cdot10^{-13}$ of the maximum error (see (20)). Bottom: Padé orders $(3,4)$–$(10,11)$ on grids selected for a 60° aperture in a distorted cell ($h_\xi = 1.1$, $h_\eta = 1.3$).

and the exact factor is $e^{i\Delta\xi(p-\beta)}$ with $p = h_\xi k\cos\theta$. The dispersion error per unit physical length is

$$\varepsilon(\theta) = \frac{\left|\log\left(\lambda e^{-i\Delta\xi(p-\beta)}\right)\right|}{h_\xi\Delta\xi}. \quad (19)$$

Locally, the transformed scheme is the flat scheme on a physical grid. Let $\beta_e = \beta/h_\xi$, $\Delta x_e = h_\xi\Delta\xi$ and $\Delta z_e = h_\eta\Delta\eta$. Then $\beta\Delta\xi = \beta_e\Delta x_e$, and the coefficients of R depend on this product only; moreover $\nu = k^2/\beta_e^2 - 1$, $\tau r = \beta_e\Delta z_e$ and $q\Delta\eta = k_z\Delta z_e$. Hence L_h and λ coincide with those of the flat scheme, and $e^{i\Delta\xi(h_\xi k\cos\theta - \beta)} = e^{i\Delta x_e(k\cos\theta - \beta_e)}$, so

$$\varepsilon(\theta; \beta, \Delta\xi, \Delta\eta, h_\xi, h_\eta) = \varepsilon_0(\theta; \beta_e, \Delta x_e, \Delta z_e), \quad (20)$$

where ε_0 is the dispersion error of the flat scheme (4) with the given reference wavenumber and steps.

Two consequences follow. First, the transformation introduces no angle restriction. A wave is resolved if its angle to the local marching direction lies within the aperture of the rational approximation on the local grid. Waves reflected by slopes of up to θ_s reach $\theta_{\max} + 2\theta_s$, so the aperture is set to that value. This requirement is physical and applies equally to a staircase.

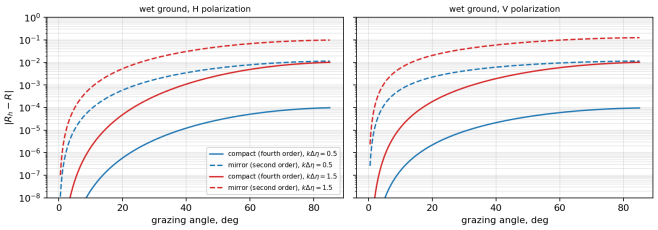


Fig. 3. Error of the discrete reflection coefficient of the impedance ground ($\varepsilon = 25 + 1.2i$, 300 MHz) for the compact closure (17) (solid) and the mirror closure (dashed), $h_\eta = 1$.

Second, the grid is selected as on flat terrain. Let $h_\pm = \max / \min h_\xi$. The flat-terrain optimizer [4], [5] is applied with:

- the wavenumber interval $[k_{\min} h_-, k_{\max} h_+]$;
- the vertical wavenumber bound $k_z^{\max} h_-$;
- the error budget per unit computational length.

The vertical step it returns is divided by $\max r \cdot h_+ / h_-$.

Fig. 2 confirms (20) numerically. It also shows that every Padé order reaches the target accuracy up to the aperture in transformed coordinates.

A. Discrete Reflection by the Impedance Ground

The same frozen-coefficient analysis applies to the ground closure.

Let a discrete wave $u_j = e^{-iqj} + R_h e^{iqj}$, $q = k \sin\phi \, h_\eta \Delta\eta$, satisfy the interior rows of (16). Row 0 then holds if and only if the ghost value $e^{iq} + R_h e^{-iq}$ satisfies (17), i.e., $2i \sin q \, (R_h - 1) + \frac{g}{3}(4 + 2\cos q)(1 + R_h) = 0$. This gives

$$R_h = \frac{iS(q) - g}{iS(q) + g}, \qquad S(q) = \frac{3 \sin q}{2 + \cos q} = q - \frac{q^5}{180} + O(q^7), \tag{21}$$

whereas the exact coefficient is $R = (iq - g)/(iq + g)$. The mirror closure gives (21) with $S = \sin q = q - q^3/6 + \ldots$. To bound the error, let $\sigma = S(q)/q$ and $w = g/(iq) = \Delta_g / \sin\phi$. Then $R = (1 - w)/(1 + w)$, $R_h = (\sigma - w)/(\sigma + w)$ and $R_h - R = 2w(\sigma - 1)/[(\sigma + w)(1 + w)]$. If $\operatorname{Re} \Delta_g \geq 0$, then $\operatorname{Re} w \geq 0$, so that $|w| \leq |1 + w|$ and $|\sigma + w| \geq \sigma$, and for both closures

$$|R_h - R| \leq \frac{2|1 - \sigma|}{\sigma} \approx \begin{cases} q^4/90, & \text{compact}, \\ q^2/3, & \text{mirror}, \end{cases} \tag{22}$$

uniformly in the ground parameters.

The condition $\operatorname{Re} \Delta_g \geq 0$ holds for any passive ground with $\operatorname{Re} \varepsilon \geq 0$, for both polarizations. The bound depends only on the vertical wavenumber resolved by the grid, as the interior dispersion error does. A grid that resolves a wave in the interior therefore also reflects it accurately, whatever the ground. Fig. 3 shows $|R_h - R|$ for wet ground. At $k\Delta\eta = 0.5$ the compact closure is accurate to $6 \cdot 10^{-5}$ up to 60°, and the mirror closure to 10^{-2}.

VIII. Compatibility With the Nonlocal Boundary Condition

Let $\eta_{\max} \geq \eta_\ell + 2\Delta\eta$, and let the medium above the transition layer be range-independent. Then the NLBC (5) computed for the chain R and the untransformed exterior scheme is the exact discrete transparent boundary condition of the scheme of Algorithm 1. Indeed, for $j\Delta\eta \geq \eta_\ell$ we have $h_\xi = h_\eta = r = m = 1$ and $V = 0$ exactly, because the taper is C^2. The rows there coincide with (4) with $z = \eta + z_{\text{ref}}$. The flux pairs are the identity in this region, and are solved below it with the top value passed through. What reaches the boundary is therefore produced by the chain R alone, with exterior coefficients independent of n, and the derivation of [6] applies verbatim, including a linear (Airy) exterior profile.

Placing the flux factors around R as in the second line of (15) is also correct, but the NLBC would then have to be computed for a chain of $p + 2q$ factors containing reciprocal pairs. The matrix function of the inverse z-transform becomes defective, and the adaptive quadrature of the kernels does not converge. The bracketed form avoids this, and the kernels are the same as over flat ground.

Above the transition layer the fine $\Delta\eta$ is needed only for the spectrum, which there is narrower than near the ground. The vertical grid can therefore be stretched, $z = Z(\eta)$, with $s = dZ/d\eta$ rising smoothly from 1 to $s_{\text{top}} = \min(\max h_\xi r, \, 2/(\bar{k}\Delta\eta))$. In the transformed equation s is a metric h_η of the Cartesian region ($m = s^2$, density weight ρs), so the scheme is unchanged. The NLBC remains exact: the z-transform, the matrix function and the convolution of [6] do not depend on the vertical grid, and only the scalar ratio u_{N-1}/u_N of the decaying exterior solution changes. On the stretched grid it satisfies a three-term recurrence with coefficients $w_j^{\mp}(1 - \alpha g_{j\mp1})$ and $-(w_j^- + w_j^+)(1 - \alpha g_j) - g_j$, where $g_j = (\beta\Delta\eta)^2 s_j^2 (\sigma - \nu_j)$ and $w_j^{\mp} = s_j / s_{j\mp1/2}$, and its minimal solution is obtained by a backward continued fraction started from the closed-form ratio of the uniform tail (step $s_{\text{top}}\Delta\eta$). With the top placed inside the stretching, this condition reproduces a taller domain to 4×10^{-6} dB for a beam reaching the top at 15°, against up to 2×10^{-2} dB for the uniform-grid condition evaluated with the local step.

IX. Comparison With Shift-Map Finite-Difference PEs

A. The Shift-Map Schemes as Rational Propagators

With $\zeta = z - T(x)$ and the gauge

$$u = v \exp\left(ik_0 T' \zeta + \tfrac{ik_0}{2} \textstyle\int_0^x T'^2 dx'\right), \tag{23}$$

the NA PE over the terrain T becomes the NA PE over flat ground for v, with the index $n^2 - 2\zeta T''$ [8], [9]. The WA shift-map PEs apply the same transformation and replace the NA operator by a WA one [10], [16]. In the FD schemes surveyed in the introduction, the envelope satisfies $u_x = ik_0(f(X) - 1)u$ with $X = k_0^{-2}\partial_\zeta^2 + n^2 - 1$, where f approximates $\sqrt{1 + X}$ by one of:

- NA: $f = 1 + X/2$;
- Claerbout: $f = (1 + \frac{3}{4}X)/(1 + \frac{1}{4}X)$;
- Greene: $f = (0.99987 + 0.79624X)/(1 + 0.30102X)$;
- Padé [2/2]: $f = (1 + \frac{5}{4}X + \frac{5}{16}X^2)/(1 + \frac{3}{4}X + \frac{1}{16}X^2)$.

The range step is Crank–Nicolson,

$$u^{n+1} = \frac{1 + i\kappa(f(X) - 1)}{1 - i\kappa(f(X) - 1)} \, u^n, \qquad \kappa = \frac{k_0 \Delta x}{2}. \tag{24}$$

X is discretized by three-point (second-order) or five-point (fourth-order) differences.

In this form the schemes are rational functions of the vertical operator, of degree $(1,1)$ or $(2,2)$, exactly like (3). The difference lies in what is approximated. Equation (24) first approximates the square root and then the exponential by the second-order Cayley transform, so its error grows with $k_0\Delta x$ and with the angle independently of the order of f. Equation (3) approximates the propagator $e^{i\beta\Delta x(\sqrt{1+L}-1)}$ itself, to order $2p$ in L. Its error vanishes uniformly as the order increases, for range steps of many wavelengths. The baseline solvers used below implement (23)–(24) in the same code as the proposed method, with the Numerov weight α set to zero. The two families therefore differ only in the approximation.

B. Grid-Independent Error of the Shift Map

For a constant slope $s = \tan\theta_s$, substituting a plane wave into the gauge (23) gives the error of the shift-map PE in the limit $\Delta x, \Delta\zeta \to 0$. A plane wave at the angle α to the horizontal has $v \propto e^{ik_0(\sin\alpha - s)\zeta}$. The PE propagates it with the longitudinal wavenumber $k_0[f(-(\sin\alpha - s)^2) - 1]$, whereas the exact wavenumber is $k_0(\cos\alpha + s\sin\alpha - 1 - s^2/2)$. Hence

$$\varepsilon_\infty(\alpha;\theta_s) = k_0\left| f\left(-(\sin\alpha - s)^2\right) + \frac{s^2}{2} - s\sin\alpha - \cos\alpha \right|, \tag{25}$$

after removal of the angle-independent part $f(0) - 1$. For $f = 1 + X/2$ this is $k_0|1 - \frac{1}{2}\sin^2\alpha - \cos\alpha|$, independent of the slope: the BT transformation is exact for the NA PE, and its floor is that of the NA PE on flat ground. For any other f, including the exact square root, the floor depends on the slope. It vanishes only on a narrow set of angles because the gauge is paraxial.

Fig. 4 shows (25) as a function of the angle to the slope. At an accuracy of $10^{-4}\,\mathrm{rad}/\lambda$ ($0.1\,\mathrm{rad}$ over $10^3\lambda$), the admissible angles are as follows.

- *Exact square root (any WA scheme):* only waves between 3.9° and 6.3° above the horizontal on a 10° slope, a single angle on a 20° slope, and none on a 30° slope.
- *Claerbout approximation:* the same windows on slopes. On these slopes the floor is set by the gauge, not by the square-root approximation, so raising the order of f cannot help.
- *NA PE:* $|\alpha| < 5.9^\circ$ for every slope.

This quantifies the published observations: WA shift maps are limited to slopes of 10–20°, can be worse than the NA PE on lee-side slopes [24], [25], and their sheared vertical wavenumber is unphysical [15]. In the proposed coordinates, by (20), the frozen-coefficient error is that of the flat scheme on a local grid, and it vanishes under refinement for every slope.

C. Maximum Angle and Cost on Flat Ground

On flat ground the shift map plays no role, and the comparison concerns the propagators alone. For every method, accuracy budget $\varepsilon_{\max}$ and maximum angle $\theta_{\max}$, we searched by brute force over $10^{-2.5} \le \Delta x/\lambda \le 10^{2.5}$ and $10^{-3} \le \Delta z/\lambda \le 1$, and over $\beta \in [0.5, 1]k_0$ for the rational scheme. The cheapest grid is the one for which the dispersion error (19), with the same Numerov or FD symbols, stays below $\varepsilon_{\max}$ for all $\theta \le \theta_{\max}$. The cost is the number of tridiagonal solves per λ^2, $p/(\Delta x\Delta z)$. The pentadiagonal systems of the five-point schemes are counted as one solve, which favors the FD methods. A phase drift common to all angles (Greene: $f(0) \neq 1$; rational: $\beta \neq k_0$) is a global phase and is removed for all methods.

TABLE I
MAXIMUM PROPAGATION ANGLE (DEGREES) MEETING AN ACCURACY BUDGET

budget, rad/λ (0.1 rad over)	10^{-3} ($10^2\lambda$)	10^{-4} ($10^3\lambda$)	10^{-5} ($10^4\lambda$)
NA-FD, CN, 3-point	10.8	6.0	3.3
WA-FD Claerbout, CN, 3-point	23.8	16.0	11.0
WA-FD Greene, CN, 5-point	11.8	3.3	1.0
WA-FD Padé [2/2] sqrt, CN, 5-point	43.8	34.8	27.8
rational Padé $(1,2)$, Numerov	43.0	38.0	27.5
rational Padé $(2,3)$, Numerov	65.3	54.5	42.8
rational Padé $(4,5)$, Numerov	79.5	73.3	66.3
rational Padé $(6,7)$, Numerov	82.5	80.0	78.0
rational Padé $(8,9)$, Numerov	84.3	82.3	80.5

Table I and Figs. 5–6 show three things.

- *FD angle limits.* The FD PEs have an angle beyond which no grid meets the budget, because the error of f does not decrease with the grid (Fig. 6). At $10^{-4}\,\mathrm{rad}/\lambda$ this angle is 6° (NA), 16° (Claerbout) and 35° (Padé $[2/2]$). The Greene approximation is a minimax fit with a relative error of about 10^{-4} over a wide range of angles; it is useful only for coarser budgets.
- *No angle limit for the rational scheme.* Its reachable angle grows with the order without bound, from 38° for $(1,2)$ to 82° for $(8,9)$, and reaches 80° even at $10^{-5}\,\mathrm{rad}/\lambda$. The search interval for Δz, not the method, ends the curves in Fig. 5.
- *Cost.* Where both are feasible, the rational scheme is far cheaper. At $10^{-4}\,\mathrm{rad}/\lambda$ the cost ratio Claerbout/rational is 130 at 10° and 560 at 15°, and the ratio Padé-$[2/2]$-FD/rational is 20, 43 and 99 at 10°, 20° and 30°. The reason is mainly the range step. The Padé approximation of the propagator is accurate to high order in $\beta\Delta x$ and allows steps of the order of the wavelength or more. The Crank–Nicolson step is only second order in $k_0\Delta x(f-1)$, so its step must shrink quickly as the angle grows.

D. Confirmation Against Exact Solutions

Beams over flat ground. Complex-source-point beams ($300\,\mathrm{MHz}$, Rayleigh parameter $50\,\mathrm{m}$, divergence 2.3°) were launched at 5°–70° over a perfectly conducting ground and propagated over $300\,\mathrm{m}$. NA-FD and WA-FD (Claerbout) were run on a typical grid ($\lambda/2 \times \lambda/10$) and on a fine grid ($\lambda/8 \times \lambda/40$). The rational scheme used the grid chosen by its optimizer for the aperture $\theta + 8^\circ$ and precision 10^{-2}. All three use the NLBC on top, so only the propagator differs. Fig. 7 shows the maximum error with respect to the exact field.

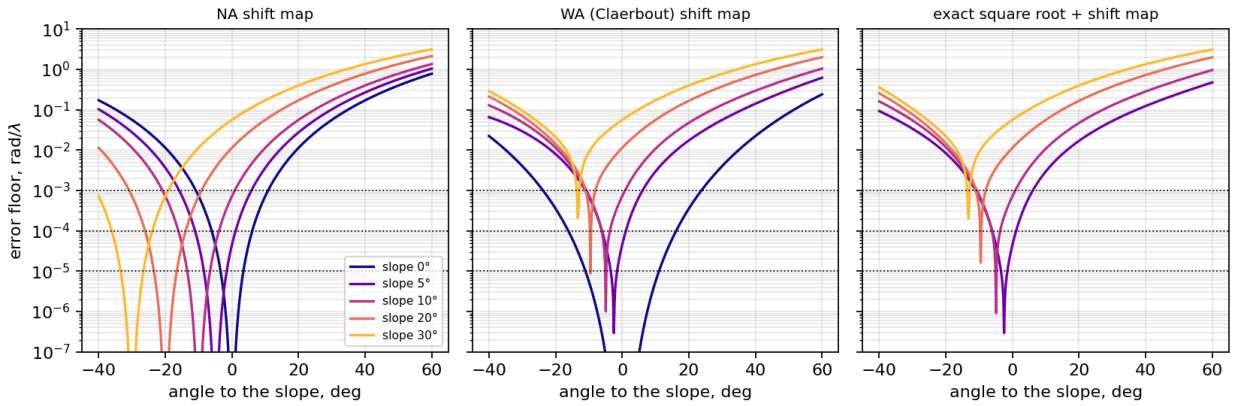


Fig. 4. Grid-independent error (25) of the shift map on a slope, versus the angle to the slope, for the NA approximation, the Claerbout approximation and the exact square root. Dotted: accuracy budgets of 10^{-3}, 10^{-4} and 10^{-5} rad/λ. For the proposed coordinates the corresponding error is zero.

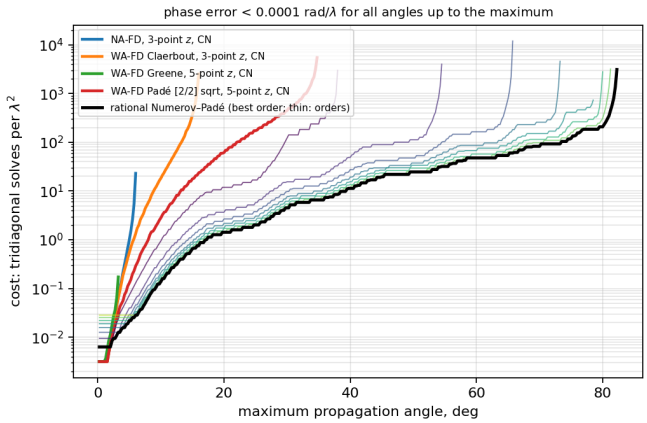


Fig. 5. Minimal cost (tridiagonal solves per λ^2) to keep the dispersion error below 10^{-4} rad/λ for all angles up to the maximum. Thick black: the rational scheme with the best order; thin lines: individual Padé orders $(1,2)$–$(8,9)$.

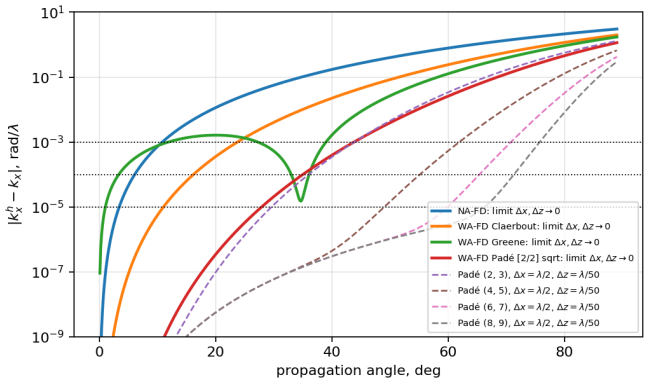


Fig. 6. Dispersion error versus angle: grid-independent limits of the FD PEs (solid) and the rational scheme on a coarse grid, $\Delta x = \lambda/2$, $\Delta z = \lambda/50$ (dashed).

- *FD PEs.* WA-FD is accurate at 5° but fails at 15° (2.6% on the fine grid) and beyond. NA-FD exceeds 3% already at 5°. Refining the grid does not help at 30° and above, in agreement with Table I.

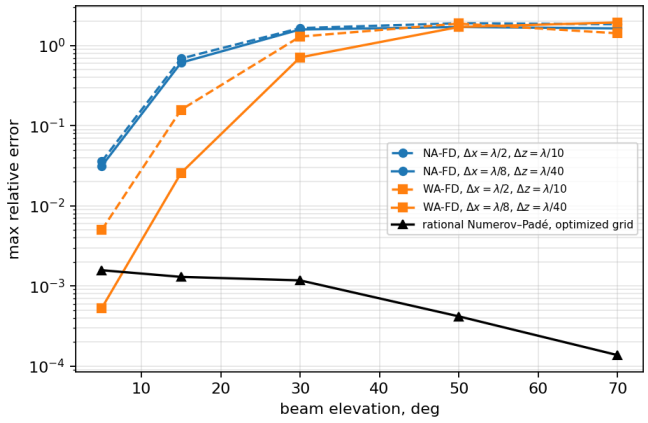


Fig. 7. Beams over flat perfectly conducting ground: maximum error with respect to the exact solution versus elevation, for NA-FD and WA-FD on two grids and for the rational scheme on its optimized grid.

- *Rational scheme.* Its error is $1.6 \cdot 10^{-3}$ to $1.4 \cdot 10^{-4}$ at all angles up to 70°. At 5° it uses 49 times fewer grid nodes than the typical FD grid, and 790 times fewer than the fine one.

Beam reflected by a slope. For the exact problem of Section XI-A the NA and WA shift maps were run on the fine grid ($\lambda/8 \times \lambda/40$). Their maximum errors are 0.67/0.56, 1.02/0.96, 1.41/1.68 and 1.19/1.29 (NA/WA) for slopes of 5°, 10°, 20° and 30°. The terrain-following rational solver gives $8.3 \cdot 10^{-5}$, $1.4 \cdot 10^{-4}$, $1.2 \cdot 10^{-3}$ and $2.9 \cdot 10^{-3}$. Already at 5° the reflected beam (20°) lies outside the admissible windows of Fig. 4.

X. Discrete NLBC Versus Absorbing Layers

Shift-map and split-step PEs truncate the domain by absorbing layers. The complex-index layer of [14] is $n^2 \to n^2 + iA\left((z - z_a)/D\right)^2$ over a layer of thickness D above the height of interest H, with $\psi = 0$ on top; window filters [27] act similarly. Such a layer works for a range of angles that depends on A and D:

- a weak or thin layer lets steep waves reach the top and reflect;

- a strong one reflects grazing waves at its gradient.

The discrete NLBC has no parameters: it is exact for the discrete scheme at every angle.

To isolate the boundary error we used the same propagator in all runs (Padé $(7,8)$, Numerov, $\Delta x = 2.5\,\mathrm{m}$, $\Delta z = 0.1\,\mathrm{m}$ at $300\,\mathrm{MHz}$), and only the top boundary was changed. On this grid the propagation error is small. With the NLBC, a domain $200\,\mathrm{m}$ taller gives the same field to all digits. The setup is:

- beams at $1°$, $3°$, $10°$ and $30°$ leave the region of interest $z < H = 200\,\mathrm{m}$ through the top;
- the ranges are 17.2, 5.7, 1.7 and $0.5\,\mathrm{km}$, so that each beam crosses H well inside the domain;
- the NLBC is placed at $H + 20\,\mathrm{m}$;
- the absorbing layers have $D = H/4$, $H/2$ and H, and $A = 10^{-3}$–1.

Fig. 8 shows the maximum error in $z < H$.

- *NLBC accuracy.* With the NLBC the error is $4 \cdot 10^{-7}$, $1.4 \cdot 10^{-6}$ and $4 \cdot 10^{-5}$ for $1°$, $3°$ and $10°$. At $30°$ it is $4.6 \cdot 10^{-3}$, the propagation error of this grid at that angle, which every boundary shares.
- *Absorbing layers.* Their errors depend strongly on the angle and on (A, D). For grazing beams no setting comes close to the NLBC. At $1°$ the best layer ($D = H$, $A = 10^{-3}$) gives $1.6 \cdot 10^{-3}$, four thousand times the NLBC error, and stronger layers give up to $8 \cdot 10^{-2}$. At $3°$ the best value is $4.7 \cdot 10^{-5}$ against $1.4 \cdot 10^{-6}$.
- *Steep beams.* Weak or thin layers reflect, with errors up to 0.28 for $D = H/4$.
- *No universal setting.* The value $A = 10^{-2}$, $D = H$ is acceptable from $3°$ upward, but gives $7.5 \cdot 10^{-3}$ at $1°$. The thinner layers recommended in the literature ($D \approx 0.3H$) are worse.

The absorbing layer therefore needs tuning to the angular content of the field, which is not known in advance. The NLBC needs none.

Cost. A layer of thickness $D = H$ doubles the number of vertical nodes; here, 4001 instead of 2201. The NLBC adds a one-time computation of the convolution kernels, and at each step a convolution with the history of the boundary values, $O(p^2 N_x)$ operations, independent of the number of vertical nodes. With the small vertical grids of this test these overheads dominate. The NLBC runs took 71, 11, 4.3 and $2.8\,\mathrm{s}$ for $1°$, $3°$, $10°$ and $30°$ ($N_x = 6876$, 2291, 682 and 209). The $D = H$ layers took 18–22, 5, 1.3 and $0.4\,\mathrm{s}$. The relative overhead decreases with the number of vertical nodes. Recursive or sum-of-exponentials approximations of the convolution [23], [43] remove its growth with N_x. The gain from the NLBC is therefore reliability and accuracy, not speed on small grids.

XI. Numerical Results

The method is implemented in JAX in the open-source library `pywaveprop` [44], as a subclass of its rational propagator. It reuses the Padé coefficients, the grid optimizer and the NLBC. Over flat ground it reproduces the existing propagator, including the NLBC, to $4 \cdot 10^{-14}$.

TABLE II
Exact Slope Test: Relative Maximum Error

		terrain-following			staircase		
θ_s	beam	Δx, m	Δz, m	error	Δx, m	Δz, m	error
$10°$	$30°$	2.5	0.035	$1.4 \cdot 10^{-4}$	2.5	0.063	0.54
$20°$	$50°$	2.5	0.019	$1.2 \cdot 10^{-3}$	2.5	0.035	0.69
$30°$	$70°$	1.0	0.009	$2.9 \cdot 10^{-3}$	1.0	0.016	0.81

A. Exact Solution: Beam Reflected by a Conducting Slope

The incident field is a $300\,\mathrm{MHz}$ complex-source-point beam, an exact solution of the Helmholtz equation. Its waist is at $(-50, 30)\,\mathrm{m}$, its Rayleigh parameter is $100\,\mathrm{m}$, and it is directed at $-10°$. Its image in the line $z = x \tan\theta_s$ gives the exact field above the Dirichlet slope; the reflected beam leaves at $2\theta_s + 10°$. Both solvers start from the exact field, use Padé $(7,8)$ and the aperture $\theta_{\max} = 10° + 2\theta_s + 5°$, and their grids are chosen by the respective optimizers with precision 10^{-2}. Table II gives the maximum error relative to the maximum of the exact field, and Fig. 9 shows the fields and the errors.

The terrain-following error is 10^{-4}–$3 \cdot 10^{-3}$ up to a reflected beam at $70°$. The staircase loses most of the reflected beam (Fig. 9), and its error decreases only slowly under refinement. For $\theta_s = 20°$ it is 0.69, 0.55 and 0.54 at $\Delta z = 0.035$, 0.013 and $0.005\,\mathrm{m}$ ($\lambda/200$), and still 0.45 when in addition Δx is reduced to $0.625\,\mathrm{m}$. The terrain-following grids have 1.2–1.4 times more vertical nodes, because of the expansion $\max r$ in the transition layer. Each step also costs $p + 4q$ instead of p solves, so the run time was 7–11 times that of the staircase, at errors two to three orders of magnitude smaller.

B. Role of the Flux-Conserving Step

For $\theta_s = 20°$, three variants of the range step give:

- Liouville rescaling alone ($q = 0$): error $6.4 \cdot 10^{-2}$, an amplitude drift along the reflected beam;
- unrotated $[2/2]$ flux factors: a spurious evanescent component grows to 0.14;
- rotated flux factors: $1.2 \cdot 10^{-3}$.

Both ingredients of Section V are therefore necessary.

C. Exact Solution: Beam Reflected by an Impedance Slope

For the impedance ground the incident beam is written as a superposition of plane waves with the Gaussian angular spectrum $\exp[-((\phi - \phi_0)/2°)^2]$, focused at $(-50, 30)\,\mathrm{m}$ and directed at $-5°$ (waist about $9\,\mathrm{m}$). Each plane wave is reflected by the slope with the coefficient $R(\phi)$ of (2), so the sum is an exact solution of the Helmholtz equation with the impedance condition on the slope. The spectrum is truncated where it falls below e^{-36}; the truncated sum is still exact. We take three grounds at $300\,\mathrm{MHz}$: wet ground ($\varepsilon = 25 + 1.2i$), dry ground ($4 + 0.006i$) and sea water ($70 + 300i$). For V polarization over wet ground the Brewster angle is $11°$, and the reflected beam is weak and sensitive to the impedance. The solver uses Padé $(7,8)$, the aperture $5° + 2\theta_s + 8°$, and the grid of its optimizer for precision 10^{-2}. Table III gives the error over the whole domain.

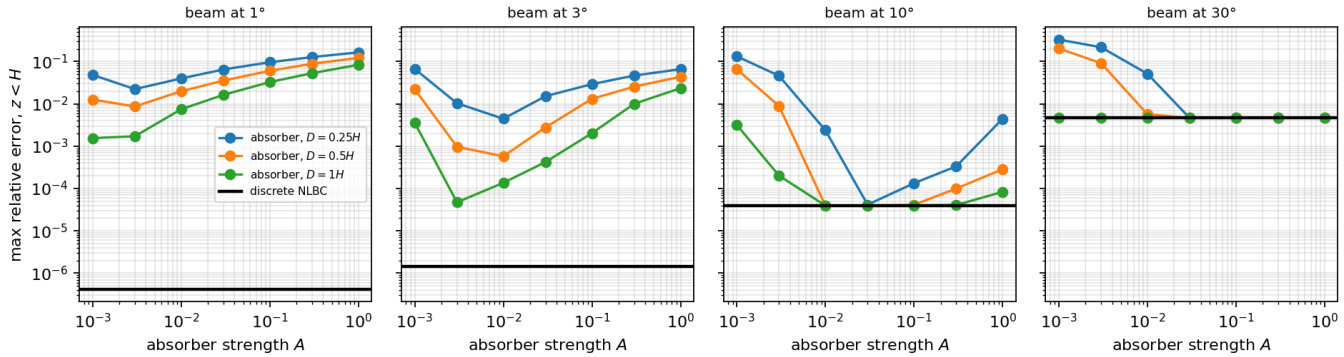


Fig. 8. Maximum error below H for beams leaving the domain through the top: absorbing layers of thickness D and strength A versus the discrete NLBC (black). Same propagator and grid in all runs.

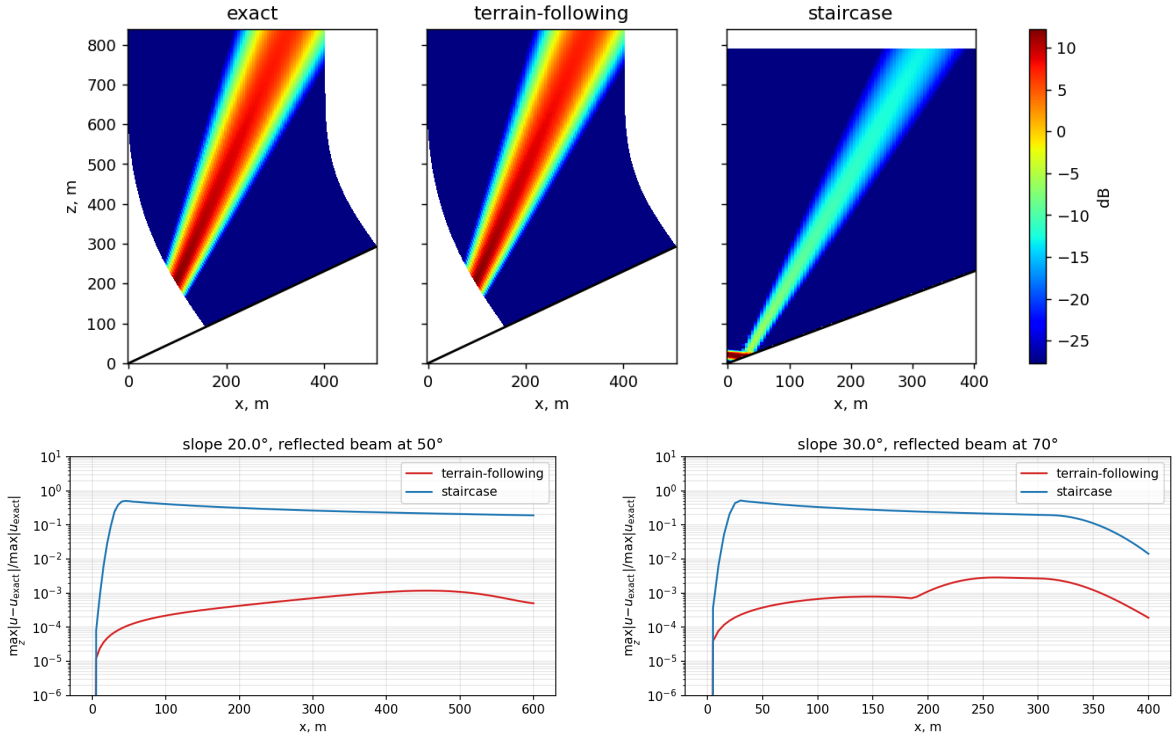


Fig. 9. Beam reflected by a perfectly conducting slope. Top: $|u|$ for $\theta_s = 30°$ (reflected beam at 70°); the terrain-following domain is bounded by the curved trajectories $\xi = 0$ and $\xi = x_{\max}$. Bottom: maximum error over height versus range for $\theta_s = 20°$ and 30°.

TABLE III
EXACT IMPEDANCE-SLOPE TEST: RELATIVE MAXIMUM ERROR

θ_s	ground	Δx, m	$\Delta\eta$, m	H	V
5°	wet	2.5	0.063	$4.4 \cdot 10^{-5}$	$3.9 \cdot 10^{-5}$
10°	wet	2.5	0.037	$9.2 \cdot 10^{-5}$	$3.0 \cdot 10^{-5}$
10°	dry	2.5	0.037	$8.2 \cdot 10^{-5}$	$3.8 \cdot 10^{-5}$
10°	sea	2.5	0.037	$9.6 \cdot 10^{-5}$	$8.1 \cdot 10^{-5}$
20°	wet	2.5	0.020	$8.1 \cdot 10^{-4}$	$8.2 \cdot 10^{-4}$
30°	wet	1.25	0.010	$9.3 \cdot 10^{-4}$	$9.2 \cdot 10^{-4}$

The error is below 10^{-3} for all grounds and both polarizations, up to a reflected beam at 65°, and is of the same size as for the conducting slope. To isolate the ground closure, the case $\theta_s = 10°$ over wet ground was repeated on prescribed grids ($\Delta x = 2.5\,\mathrm{m}$, $\beta = k\cos 20°$) with both closures:

$k\Delta\eta$	0.39	0.63	1.05	1.57
V, compact	$7.5 \cdot 10^{-5}$	$2.5 \cdot 10^{-4}$	$1.7 \cdot 10^{-3}$	$8.6 \cdot 10^{-3}$
V, mirror	$7.6 \cdot 10^{-4}$	$1.9 \cdot 10^{-3}$	$5.4 \cdot 10^{-3}$	$1.3 \cdot 10^{-2}$
H, both	$2.6 \cdot 10^{-4}$	$8.1 \cdot 10^{-4}$	$5.4 \cdot 10^{-3}$	$2.7 \cdot 10^{-2}$

For V polarization the mirror closure dominates the error on fine grids, and the compact closure reduces it by a factor of 3–10. For H polarization $|w| = |\Delta_g|/\sin\phi \approx 20$, so $|R_h - R| \approx 2|1-\sigma|/|w|$ is small for both closures, and the error is that of the interior scheme. The compact closure costs nothing extra, since the system remains tridiagonal.

TABLE IV
IMPEDANCE SLOPES, BEAM AT $-5°$: RELATIVE MAXIMUM ERROR

	terrain-following	staircase			
θ_s	all grounds	wet, H	wet, V	sea, V	conducting
$2°$	$3.6 \cdot 10^{-5}$	0.026	0.061	0.16	0.041
$5°$	$4.7 \cdot 10^{-5}$	0.068	0.15	0.28	0.13
$10°$	$9.2 \cdot 10^{-5}$	0.21	0.27	0.39	0.30
$20°$	$8.2 \cdot 10^{-4}$	0.54	0.30	0.45	0.90

D. Staircase over Gentle Impedance Slopes

The conducting-slope test of Section XI-A involves steep beams. To check whether the staircase is adequate in the regime of most radio paths, grazing incidence on gentle slopes over real ground, the impedance-slope problem of Section XI-C was solved with slopes $\theta_s = 2$–$20°$ by a classical staircase solver. It uses Padé $[8/8]$ steps with $\beta = k$, the Numerov scheme, $\Delta x = 1\,\mathrm{m}$ and $\Delta z = 0.025\,\mathrm{m}$ ($\lambda/40$), sets the field to zero below the ground node of each column, and imposes the Leontovich condition $u_z + \alpha u = 0$ at that node with the mirror closure, i.e., with the vertical normal of the step. On flat ground it reproduces the exact solution to $1.2 \cdot 10^{-5}$. Table IV compares it with the terrain-following solver of Section XI-C.

Already at $\theta_s = 2°$ the staircase error is 3–16%, and it grows with the slope; the terrain-following error stays below 10^{-4} up to $10°$. Refinement does not help: for $\theta_s = 5°$, wet ground and V polarization, the error is 0.153 at $\Delta z = 0.025\,\mathrm{m}$, 0.153 at $0.0125\,\mathrm{m}$, and 0.155 when in addition Δx is halved. The error is systematic, not numerical noise. The steps impose the reflection coefficient of the grazing angle to the horizontal instead of to the slope. In Fig. 10 the beam meets the slope at $10°$, near the Brewster angle of $11°$, where $|R| = 0.06$, whereas the steps reflect it with $|R(5°)| = 0.38$: the reflected beam is $9\,\mathrm{dB}$ too strong. Over sea water the pseudo-Brewster angle lies at a few degrees, so even a $2°$ slope changes $|R|$ by about 20% and the staircase error is 16%. For $\theta_s \le 5°$ the staircase errs more for V polarization and for sea water than over a perfect conductor, whose reflection does not depend on the angle, so conducting benchmarks understate the staircase error over real ground at grazing incidence.

E. Real Terrain and NWP Refractivity

A $300\,\mathrm{MHz}$ horizontally polarized antenna on a $30\,\mathrm{m}$ mast stands on top of a ridge in the Scottish Highlands ($57.2163°$N $4.9214°$W) and radiates along a $28\,\mathrm{km}$ path to $57.40°$N $4.60°$W. Its beam is $4°$ wide and tilted $1°$ down.

- *Terrain:* SRTM-type elevation at $0.0005°$, from 166 to $664\,\mathrm{m}$, smoothed with $\sigma = 100\,\mathrm{m}$; the maximum slope is $11.1°$.
- *Refractivity:* the modified refractivity of the GFS analysis of 2026-10-09 12 UTC, interpolated along the path and averaged over range.
- *Ground:* perfectly conducting.
- *Grid parameters:* aperture $5° + 2 \cdot 11.1°$, transition height $L = 1000\,\mathrm{m}$, $z_{\mathrm{ref}} = \overline{T}$.

Both solvers see the same smoothed terrain. Fig. 11 shows the path loss.

Each solver is converged with respect to its own grid. The two precision settings (10^{-2} and 10^{-3}) differ, at $10\,\mathrm{m}$ above the ground, by $0.12\,\mathrm{dB}$ (median) for the terrain-following solver and by $0.15\,\mathrm{dB}$ for the staircase. The two solvers differ by $1.1\,\mathrm{dB}$ at $10\,\mathrm{m}$ and $0.43\,\mathrm{dB}$ at $50\,\mathrm{m}$ above the ground. The largest differences, up to a few decibels, occur on the slopes facing the transmitter, where the staircase solution carries small-scale ripple consistent with scattering by the steps (Section XI-A). Run alone on a CPU, the terrain-following solver takes 298 and 787 s, and the staircase 50 and 80 s. The terrain-following grid ($\Delta x = 3.1\,\mathrm{m}$, $\Delta\eta = 0.033$ and $0.018\,\mathrm{m}$) is finer than the staircase grid ($\Delta x = 12.5$ and $10\,\mathrm{m}$, $\Delta z = 0.069$ and $0.038\,\mathrm{m}$) because of the metric spread $h_\xi \in [0.91, 1.21]$ and the expansion $\max r = 1.26$, which enter the grid selection through the global bounds of the grid selection rule of Section VII. With one reference wavenumber per class of columns and an absorbing layer on top (Section XI-F) the runs take 139 and 367 s, and the path loss changes by 0.003 and $0.04\,\mathrm{dB}$ (median); the staircase remains 2.8 and 4.6 times cheaper.

F. Steep Terrain over Real Ground

The Highlands path has a perfectly conducting ground and slopes of $11°$. To see whether the effect of Section XI-D survives on a real path, and how it compares with environmental uncertainty, a steeper path with real ground was computed. The transmitter stands on the floor of Glen Etive, Scottish Highlands ($56.6064°$N $4.9986°$W, $53\,\mathrm{m}$), and radiates along a $5.3\,\mathrm{km}$ path to $56.645°$N $5.05°$W, up a mountain flank that rises to $750\,\mathrm{m}$.

- *Source:* $100\,\mathrm{MHz}$, $30\,\mathrm{m}$ mast, beam $10°$ wide and tilted $3°$ up, H and V polarization.
- *Terrain:* smoothed with $\sigma = 100\,\mathrm{m}$; the maximum slope is $23.5°$.
- *Ground:* wet ground, $\varepsilon = 25$, $\sigma = 0.02\,\mathrm{S/m}$, whose Brewster angle for V polarization is about $11°$.
- *Atmosphere:* standard, $dM/dz = 0.118\,\mathrm{M/m}$.

The staircase is the solver of Section XI-D: the Leontovich condition is imposed at the ground node of every column, and the rational coefficients, β, Δx and Δz are those chosen by the grid optimizer for the same precision. Its aperture is $10° + 3° + 2 \cdot 23.5° = 60°$, since rays reflected by the slope leave at twice the slope from the horizontal. Near the ground the same rays stay within one slope of the terrain-following coordinate lines, so the terrain-following solver uses $10° + 3° + 23.5° = 36.5°$. A terrain-following run with the full $60°$ differs from it by at most $0.05\,\mathrm{dB}$ along the cuts, and its grid is ten times larger. At precisions 10^{-2} and 10^{-3} the terrain-following grid has $\Delta\xi = 5\,\mathrm{m}$, $\Delta\eta = 0.17$ and $0.09\,\mathrm{m}$, and 101 and 212 million node solves (nodes times rational factors per step); the staircase, with its wider aperture, has $\Delta x = 6.25\,\mathrm{m}$, $\Delta z = 0.125$ and $0.077\,\mathrm{m}$, and 140 and 260 million. Run alone on a CPU, the terrain-following solver takes 18 and 29 s, and the package staircase solver on the staircase grid 15 and 22 s;

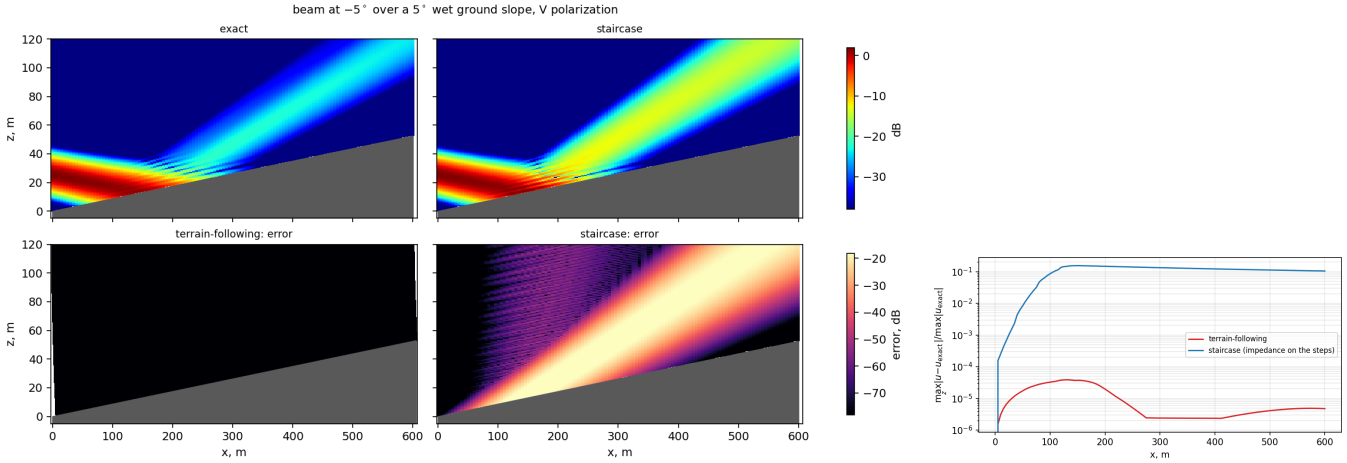


Fig. 10. Beam at -5° over a 5° wet-ground slope, V polarization. Left: exact field, staircase field, and the errors of both solvers. Right: maximum error over height versus range.

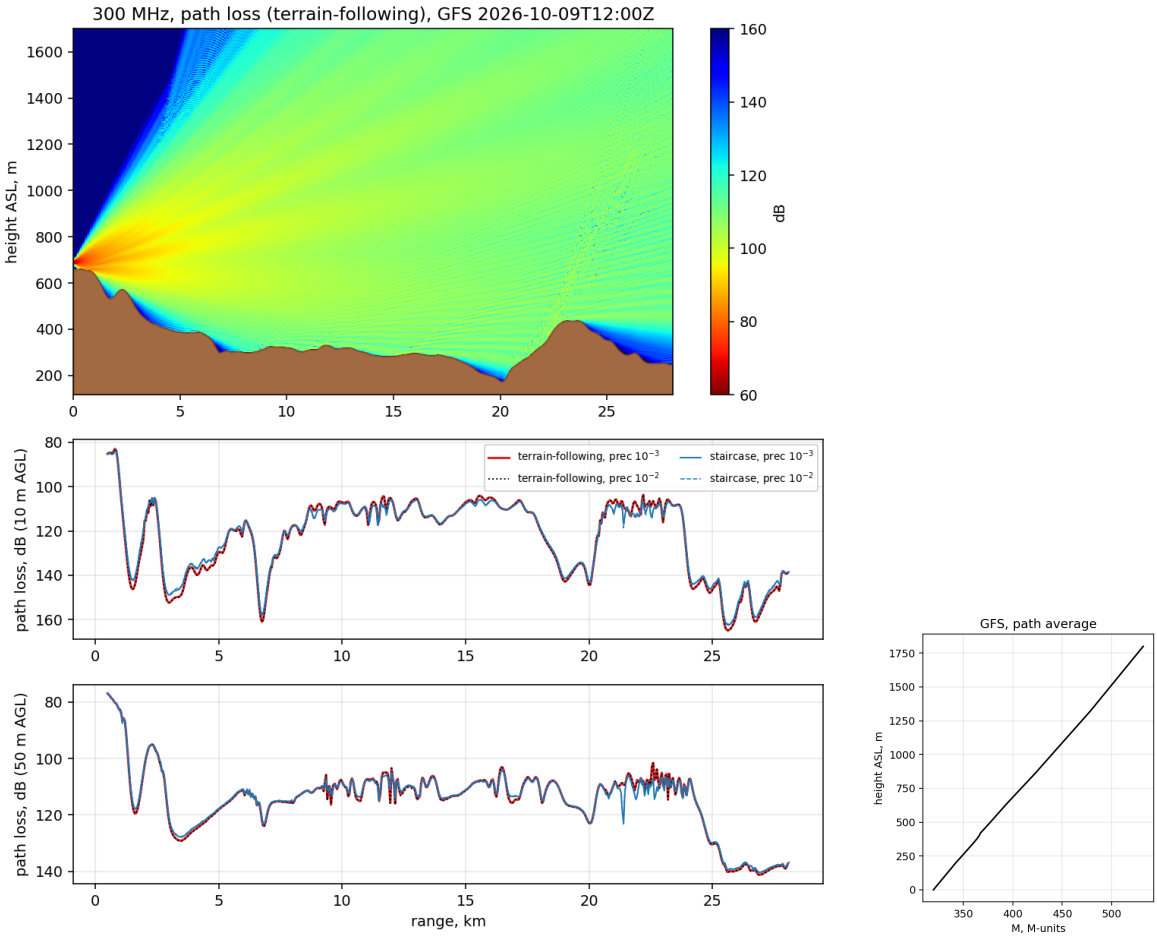


Fig. 11. Scottish Highlands, 300 MHz. Left: path loss of the terrain-following solver, and path loss 10 m and 50 m above the ground for both solvers at two precision settings. Right: GFS modified refractivity averaged along the path.

the terrain-following step also applies the flux factors, which are not counted in the node solves.

The reference wavenumber β of the terrain-following run has to cover the local spectrum kh_ξ of every column. With one β per class of columns, grouped by the range of h_ξ on the column, the range step grows to 8.3 m at both precisions. The discrete NLBC of Section VIII assumes the same step in every column, so the upper boundary is then an absorbing layer (300 m). Its grid is stretched, $dz/d\eta$ rising smoothly to $1/(k_{z,\max}\Delta\eta)$, so that it takes 456 rows instead of 1800

TABLE V
GLEN ETIVE: PATH-LOSS DIFFERENCES (DB), V POLARIZATION, MEDIAN $|\Delta|$ / MEAN Δ

height above ground	0–100 m	100–500 m	500–2000 m
staircase − terrain-following	0.9 / −0.7	4.2 / −4.6	9.4 / −10.7
medium − wet ground	0.5 / −0.4	4.0 / −2.6	5.6 / −3.9
$dM/dz \pm 0.039\,\mathrm{M/m}$	0.01 / 0.00	0.03 / 0.00	0.03 / 0.00

at 10^{-2}. In the transformed equation the stretching is a metric $h_\eta = dz/d\eta$ of the Cartesian region, so the scheme is unchanged. This run takes 7 and 10 s, and its path loss differs from the one with a single β by $0.007\,\mathrm{dB}$ (median) and $0.02\,\mathrm{dB}$ at most.

Two checks support the staircase used here. With the masking of the package solver (no ground condition inside the step, field set to zero below the ground after it) the same code reproduces the package staircase to $0.01\,\mathrm{dB}$ (median). On a perfectly conducting ground, the staircase with the condition imposed at the ground node differs from the terrain-following solution by $0.5\,\mathrm{dB}$ (median, $10\,\mathrm{m}$ above the ground), and the package staircase by $2.1\,\mathrm{dB}$. The staircase compared below is therefore the more accurate of the two.

Fig. 12 and Table V show the result. The terrain-following solution is converged: the precisions 10^{-2} and 10^{-3} agree to better than $0.01\,\mathrm{dB}$ along the cuts. The staircase is converged with respect to its own grid as well: its two precisions differ by $0.02\,\mathrm{dB}$ (median). The difference between the solvers is therefore systematic. In H polarization it is small, at most $1.6\,\mathrm{dB}$ (median) at any height, because $|R_H|$ of wet ground varies only from 0.98 to 0.84 between 3° and 25°. In V polarization it grows with height. Close to the ground, where the direct field dominates, it is $0.9\,\mathrm{dB}$, comparable to the effect of replacing wet by medium ground. In the fan of rays reflected by the flank ($500\,\mathrm{m}$ to $2\,\mathrm{km}$ above the ground) the staircase field is $10.7\,\mathrm{dB}$ too strong on average, twice the effect of the ground parameters, and the error persists under refinement. The cause is that of Section XI-D. A ray tilted 3° up meets a 20° slope at 23°, where $|R_V| = 0.33$, whereas the steps reflect it at 3°, where $|R_V| = 0.58$. This accounts for about $5\,\mathrm{dB}$. The rest is consistent with the rays closer to the horizontal, for which the steps see $|R_V|$ approaching one. The steps also add spurious steep rays above the first slope (Fig. 12, top right). The refractivity gradient matters little on a $5\,\mathrm{km}$ path: its uncertainty changes the path loss by less than $0.1\,\mathrm{dB}$.

The comparison qualifies the conclusion drawn from the Highlands path. At receivers near the ground the staircase error is comparable to the effect of the ground parameters, which are rarely known better than the two grounds used here. For elevated receivers over steep terrain, such as aircraft, unmanned aerial vehicles or a radar looking up the slope, it is about twice that effect, and the error is a bias of one sign: the staircase reflects the V beam as if the ground were a better conductor. Unlike the effect of the ground parameters, this bias cannot be reduced by better data.

XII. DISCUSSION AND LIMITATIONS

- *When the staircase suffices.* Over smoothed terrain with a perfectly conducting ground (Section XI-E) the two solvers differ by about $1\,\mathrm{dB}$ at $10\,\mathrm{m}$ above the ground, which is below the usual uncertainty of refractivity, elevation data and ground parameters, and the staircase is three to ten times cheaper. Near the ground of a steep path over wet ground the difference is still comparable to the effect of the ground parameters (Section XI-F). For such coverage estimates the staircase is adequate. The terrain-following solver matters where the staircase error is of order one or systematic: steep slopes and wide-angle reflections (Section XI-A), and real ground, where the steps impose the reflection coefficient of the wrong grazing angle (Section XI-D). On a real mountain path this makes the field reflected by the slope $10\,\mathrm{dB}$ too strong in V polarization, twice the effect of the ground parameters, for elevated receivers (Section XI-F). That error is not reduced by refinement, it is tied to the terrain and therefore the same in every run, and it does not average out over an ensemble of environments as their uncertainty does. A method whose error stalls under refinement also cannot serve as a reference: without a convergent solver one cannot show that the numerical error is smaller than the environmental one.
- *Dependence on the terrain.* In the staircase the field depends on the terrain height through the index of the ground node, i.e., discontinuously, so its derivatives with respect to the terrain or to ground parameters on the steps are not defined. In the proposed coordinates the terrain enters through smooth coefficient tables, and the solution depends smoothly on it. This is a prerequisite for gradient-based inversion and sensitivity analysis. The propagator is written in JAX, but the map is at present computed in NumPy, so derivatives with respect to the terrain are not yet available.
- *Cost.* The grid is chosen with the extreme values of h_ξ and r over the whole domain. On the path of Section XI-F, for the same aperture, this reduces the range step from $15.9\,\mathrm{m}$ on flat ground to $5.5\,\mathrm{m}$ and the vertical step by a factor of 2. One reference wavenumber per class of columns recovers part of the range step ($8.3\,\mathrm{m}$); the vertical factor comes from the stretching of the map and remains. The aperture, in contrast, can be smaller than for the staircase: angles are measured from the coordinate lines, which near the ground follow the slope. There one slope instead of two suffices and makes the grid ten times smaller. With both, the terrain-following run takes about half as long as the staircase (7 s against 15 s at 10^{-2}).
- *Smoothness.* The terrain must be C^2. Knife edges, cliffs and overhangs are smoothed over the scale σ, and very steep relief requires a larger transition height.
- *Ground impedance.* The Leontovich condition is local and angle independent. It is accurate for $|\varepsilon| \gg 1$, but not for dry ground at steep incidence. An exact angle-dependent lower NLBC [7] over terrain would require a kernel for range-dependent boundary coefficients, which is an open

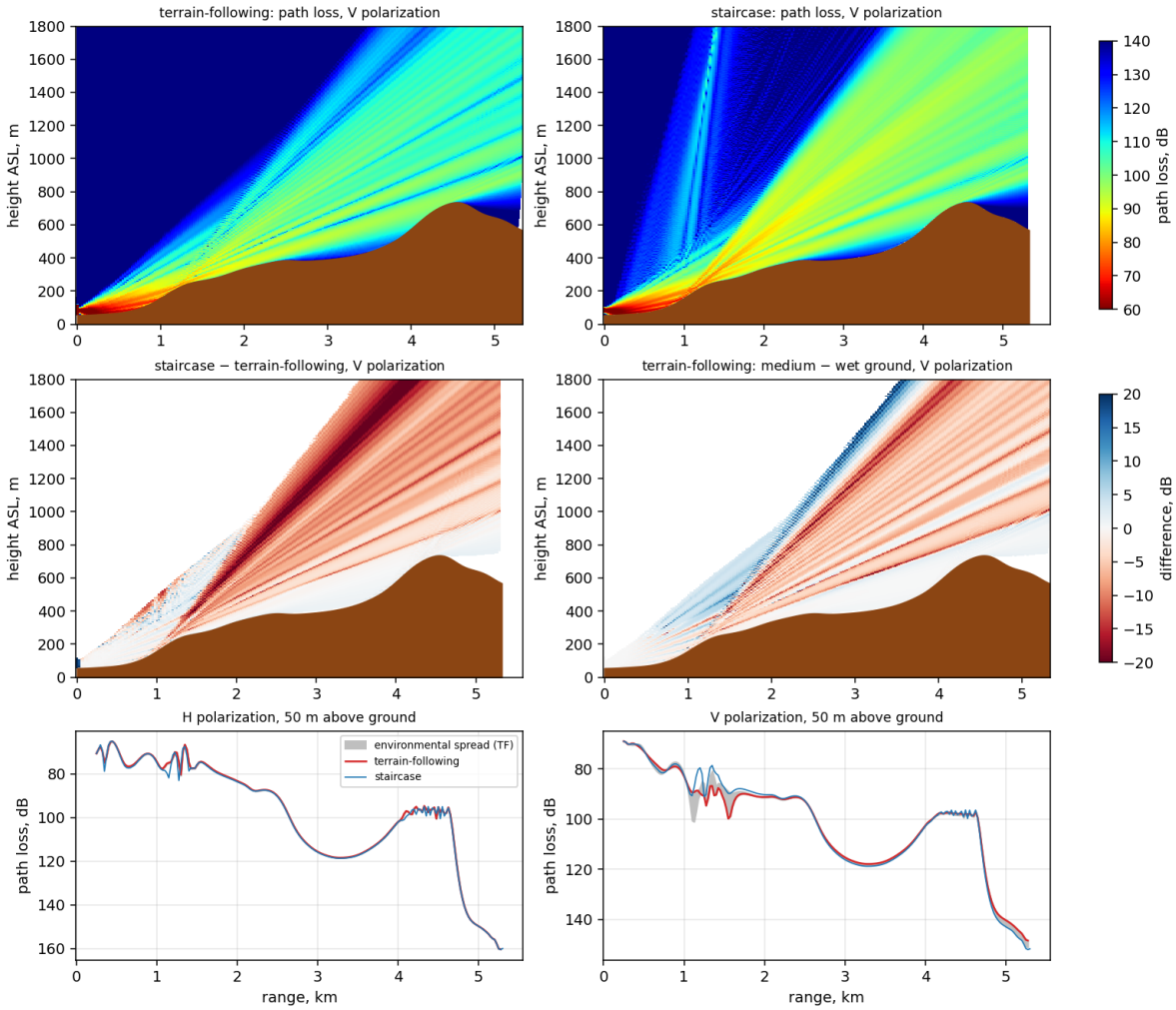


Fig. 12. Glen Etive, 100 MHz, wet ground. Top: path loss of the terrain-following and staircase solvers, V polarization. Middle: the staircase minus the terrain-following path loss, and the terrain-following path loss for medium ground ($\varepsilon = 15$, $\sigma = 0.005\,\mathrm{S/m}$) minus that for wet ground, on the same scale; the differences are shown where the path loss is below 130 dB. Bottom: path loss 50 m above the ground; the grey band is the spread of the terrain-following solution over the two grounds and $dM/dz = 0.118 \pm 0.039\,\mathrm{M/m}$.

problem.

- *Refractivity.* The medium above the transition layer must not depend on range, as for any discrete NLBC. A range-dependent medium inside the layer is admissible, since the coefficient tables are per column.

XIII. Conclusion

Orthogonal terrain-following coordinates, built from the tapered harmonic extension of the terrain and its orthogonal trajectories, turn irregular terrain into a variable coefficient and a weight in the vertical Numerov scheme. Rational approximations of arbitrary order then apply with the angle limit of the flat scheme on a local effective grid, and the discrete NLBC is used without modification. A flux-conserving step with rotated-branch-cut factors removes the amplitude drift caused by the rotation of the coordinate lines. On an exact wide-angle benchmark the method is two to three orders of magnitude more accurate than the staircase. A compact closure of the Leontovich condition reflects every resolved wave with an error bounded uniformly in the ground parameters, and keeps the error below 10^{-3} over impedance slopes for both polarizations, where a staircase errs by several percent already on gentle slopes. On a mountain path over wet ground the staircase overestimates the field reflected by the slope by 10 dB, which is more than the uncertainty of the ground parameters. Compared with the shift-map finite-difference PEs, it has no grid-independent slope error and no maximum propagation angle, and it costs one to three orders of magnitude less at moderate angles. Unlike absorbing layers, the discrete transparent boundary needs no tuning and is accurate for grazing as well as steep waves.

Code Availability

The implementation, tests and the scripts reproducing all figures are available in the `pywaveprop` repository [44].

References

[1] M. Levy, *Parabolic Equation Methods for Electromagnetic Wave Propagation*, ser. IEE Electromagnetic Waves Series 45. London: IEE, 2000.
[2] M. D. Collins, "A split-step Padé solution for the parabolic equation method," *J. Acoust. Soc. Am.*, vol. 93, no. 4, pp. 1736–1742, 1993.

[3] M. S. Lytaev, "Numerov-Padé scheme for the one-way Helmholtz equation in tropospheric radio-wave propagation," *IEEE Antennas and Wireless Propagation Letters*, vol. 19, no. 12, pp. 2167–2171, 2020.

[4] ——, "Automated selection of the computational parameters for the higher-order parabolic equation numerical methods," in *Computational Science and Its Applications – ICCSA 2020*, ser. Lecture Notes in Computer Science, vol. 12249. Springer, 2020, pp. 296–311.

[5] ——, "Mesh optimization for rational approximations of the one-way Helmholtz equation in inhomogeneous media," *Numerical Methods and Programming (Vychislitel'nye Metody i Programmirovanie)*, vol. 27, no. 2, pp. 302–316, 2026.

[6] ——, "Nonlocal boundary conditions for split-step Padé approximations of the Helmholtz equation with modified refractive index," *IEEE Antennas and Wireless Propagation Letters*, vol. 17, no. 8, pp. 1561–1565, 2018.

[7] ——, "Fresnel reflection modeling within the higher-order parabolic equation and discrete nonlocal boundary conditions," in *2022 IEEE Radar Conference (RadarConf22)*, 2022.

[8] A. Beilis and F. D. Tappert, "Coupled mode analysis of multiple rough surface scattering," *The Journal of the Acoustical Society of America*, vol. 66, no. 3, pp. 811–826, 1979.

[9] A. E. Barrios, "A terrain parabolic equation model for propagation in the troposphere," *IEEE Transactions on Antennas and Propagation*, vol. 42, no. 1, pp. 90–98, 1994.

[10] D. J. Donohue and J. R. Kuttler, "Propagation modeling over terrain using the parabolic wave equation," *IEEE Transactions on Antennas and Propagation*, vol. 48, no. 2, pp. 260–277, 2000.

[11] A. E. Barrios, "Considerations in the development of the advanced propagation model (APM) for U.S. Navy applications," in *Proceedings of the International Conference on Radar (RADAR 2003)*, 2003, pp. 77–82.

[12] L. B. Dozier, "PERUSE: A numerical treatment of rough surface scattering for the parabolic wave equation," *The Journal of the Acoustical Society of America*, vol. 75, no. 5, pp. 1415–1432, 1984.

[13] S. Parakkal, K. E. Gilbert, and X. Di, "Application of the Beilis–Tappert parabolic equation method to sound propagation over irregular terrain," *The Journal of the Acoustical Society of America*, vol. 131, no. 2, pp. 1039–1046, 2012.

[14] C. Khodr, M. Azarpeyvand, and D. N. Green, "An iterative three-dimensional parabolic equation solver for propagation above irregular boundaries," *The Journal of the Acoustical Society of America*, vol. 148, no. 2, pp. 1089–1100, 2020.

[15] K. E. Gilbert, "Wide-angle formulation of the Beilis-Tappert method," *The Journal of the Acoustical Society of America*, vol. 152, no. 2, pp. 1170–1179, 2022.

[16] P. D. Holm, "Wide-angle shift-map PE for a piecewise linear terrain—a finite-difference approach," *IEEE Transactions on Antennas and Propagation*, vol. 55, no. 10, pp. 2773–2789, 2007.

[17] C. Zhou, Q. Zeng, W. Ding, and Y. Long, "Greene second-order approximation of the wide-angle equation for terrain," in *2015 Asia-Pacific Microwave Conference (APMC)*, 2015.

[18] Q. Guo, C. Zhou, and Y. Long, "Greene approximation wide-angle parabolic equation for radio propagation," *IEEE Transactions on Antennas and Propagation*, vol. 65, no. 11, pp. 6048–6056, 2017.

[19] Q. Guo and Y. Long, "Padé second-order parabolic equation modeling for propagation over irregular terrain," *IEEE Antennas and Wireless Propagation Letters*, vol. 16, 2017.

[20] Q. Guo, Y. Lan, X. Deng, and Y. Diao, "Application of the finite difference parabolic equation model in forestry remote telemetry," *International Journal of Antennas and Propagation*, vol. 2023, p. 6825971, 2023.

[21] N. Zhang and Y. Long, "New difference schemes for impedance boundary conditions of wide-angle shift-map PE," *IEEE Transactions on Antennas and Propagation*, vol. 71, no. 10, pp. 8378–8383, 2023.

[22] N. Zhang, Z. Liang, and Y. Long, "An efficient difference scheme for wide-angle shift-map parabolic equation," *IEEE Transactions on Antennas and Propagation*, vol. 72, no. 4, pp. 3545–3552, 2024.

[23] R. Wang, G. Lu, R. Zhang, and W. Xu, "Combination of the improved diffraction nonlocal boundary condition and three-dimensional wide-angle parabolic equation decomposition model for predicting radio wave propagation," *International Journal of Antennas and Propagation*, vol. 2017, p. 2728380, 2017.

[24] D.-D. Wang, Y.-R. Pu, X.-L. Xi, and L.-L. Zhou, "An analysis of narrow-angle and wide-angle shift-map parabolic equations," *IEEE Transactions on Antennas and Propagation*, vol. 68, no. 5, pp. 3911–3918, 2020.

[25] L. Zhou, D. Wang, Z. Mu, X. Xi, and L. He, "LF radio wave prediction at short ranges with high propagation angles over irregular terrain," *IEEE Antennas and Wireless Propagation Letters*, vol. 16, pp. 732–735, 2017.

[26] R. Zhang, H. Wang, Z. Wu, L. Lin, and Z. Zhao, "Prediction method of maximum propagation angle in parabolic equation model over irregular terrain," *International Journal of Antennas and Propagation*, vol. 2024, p. 4988555, 2024.

[27] P. Zhang, L. Bai, Z. Wu, and F. Li, "Effect of window function on absorbing layers top boundary in parabolic equation," in *Proceedings of 2014 3rd Asia-Pacific Conference on Antennas and Propagation (APCAP)*, 2014, pp. 849–852.

[28] M. F. Levy, "Perfectly matched layer truncation for parabolic wave equation models," *Proceedings of the Royal Society of London. Series A*, vol. 457, no. 2015, pp. 2609–2624, 2001.

[29] S. Ozbayat and R. Janaswamy, "Effective local absorbing boundary conditions for a finite difference implementation of the parabolic equation," *IEEE Transactions on Antennas and Propagation*, vol. 59, no. 5, pp. 1616–1625, 2011.

[30] X. Zhang and C. D. Sarris, "Error analysis and comparative study of numerical methods for the parabolic equation applied to tunnel propagation modeling," *IEEE Transactions on Antennas and Propagation*, vol. 63, no. 7, pp. 3025–3034, 2015.

[31] M. D. Collins, "The rotated parabolic equation and sloping ocean bottoms," *J. Acoust. Soc. Am.*, vol. 87, no. 3, pp. 1035–1037, 1990.

[32] D. A. Outing, W. L. Siegmann, M. D. Collins, and E. K. Westwood, "Generalization of the rotated parabolic equation to variable slopes," *J. Acoust. Soc. Am.*, vol. 120, no. 6, pp. 3534–3538, 2006.

[33] D. J. Donohue and J. R. Kuttler, "Modeling radar propagation over terrain," *Johns Hopkins APL Technical Digest*, vol. 18, no. 2, pp. 279–287, 1997. [Online]. Available: https://secwww.jhuapl.edu/techdigest/Content/techdigest/pdf/V18-N02/18-02-Donohue.pdf

[34] M. Heiblum and J. H. Harris, "Analysis of curved optical waveguides by conformal transformation," *IEEE Journal of Quantum Electronics*, vol. 11, no. 2, pp. 75–83, 1975.

[35] J. T. Kirby, R. A. Dalrymple, and H. Kaku, "Parabolic approximations for water waves in conformal coordinate systems," *Coastal Engineering*, vol. 23, no. 3-4, pp. 185–213, 1994.

[36] Y. Y. Lu, J. Huang, and J. R. McLaughlin, "Local orthogonal transformation and one-way methods for acoustic waveguides," *Wave Motion*, vol. 34, no. 2, pp. 193–207, 2001.

[37] N. A. Kampanis, "A finite element method for the parabolic equation in aeroacoustics coupled with a nonlocal boundary condition for an inhomogeneous atmosphere," *Journal of Computational Acoustics*, vol. 13, no. 4, pp. 569–584, 2005.

[38] M. D. Collins and E. K. Westwood, "A higher-order energy-conserving parabolic equation for range-dependent ocean depth, sound speed, and density," *J. Acoust. Soc. Am.*, vol. 89, no. 3, pp. 1068–1075, 1991.

[39] M. B. Porter, F. B. Jensen, and C. M. Ferla, "The problem of energy conservation in one-way models," *J. Acoust. Soc. Am.*, vol. 89, no. 3, pp. 1058–1067, 1991.

[40] F. A. Milinazzo, C. A. Zala, and G. H. Brooke, "Rational square-root approximations for parabolic equation algorithms," *The Journal of the Acoustical Society of America*, vol. 101, no. 2, pp. 760–766, 1997.

[41] S. K. Lele, "Compact finite difference schemes with spectral-like resolution," *Journal of Computational Physics*, vol. 103, no. 1, pp. 16–42, 1992.

[42] G. D. Dockery and J. R. Kuttler, "An improved impedance-boundary algorithm for Fourier split-step solutions of the parabolic wave equation," *IEEE Transactions on Antennas and Propagation*, vol. 44, no. 12, pp. 1592–1599, 1996.

[43] A. Arnold and M. Ehrhardt, "Discrete transparent boundary conditions for wide angle parabolic equations in underwater acoustics," *Journal of Computational Physics*, vol. 145, no. 2, pp. 611–638, 1998.

[44] M. S. Lytaev, "PyWaveProp: wave propagation framework for Python," https://github.com/mikelytaev/wave-propagation, 2026.